Experimentos do Trabalho 1 - Reconhecimento de Padrões

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1. Imports e Carregando Dataset

In [10]:
```
import time
import warnings

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from scipy import stats
from sklearn.decomposition import PCA
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from ucimlrepo import fetch_ucirepo

from classificao_padroes.models_trabalho1 import KNN, MAXCO, MDC

warnings.filterwarnings('ignore')
```

In [11]:
```
# Carregar o dataset
vertebral_column = fetch_ucirepo(id=212)
X = vertebral_column.data.features.to_numpy()
y_raw = vertebral_column.data.targets.to_numpy().ravel()

# Codificar as labels para valores numéricos
le = LabelEncoder()
y = le.fit_transform(y_raw)

print("Shape of X:", X.shape)
print("Shape of y:", y.shape)
print("Classes:", le.classes_)
print("Encoded Classes:", np.unique(y))
```

```
Shape of X: (310, 6)
Shape of y: (310,)
Classes: ['Hernia' 'Normal' 'Spondylolisthesis']
Encoded Classes: [0 1 2]
```

1.2 Análise dos Dados

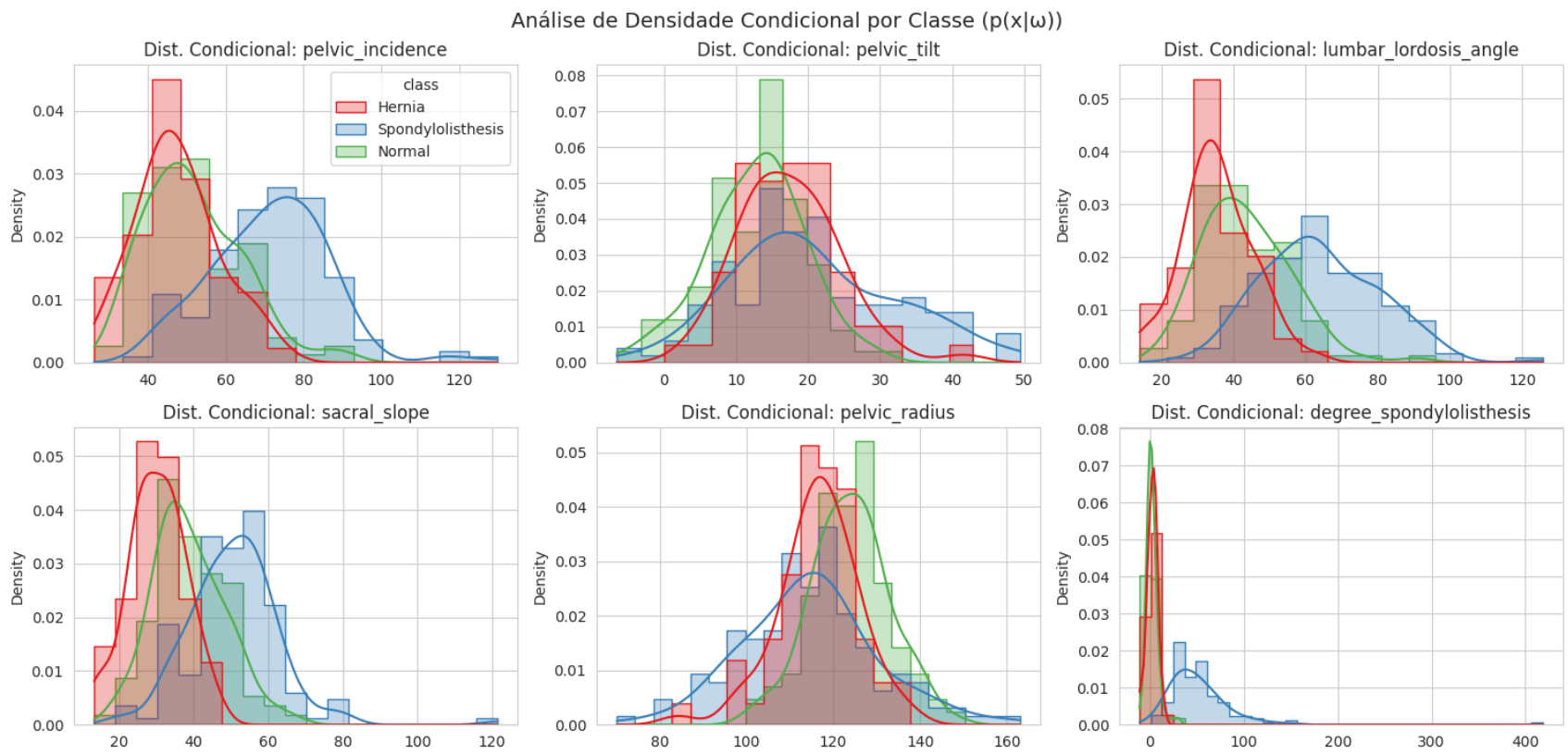
In [12]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import math

# Configurações estéticas para publicação
sns.set_style("whitegrid")
plt.rcParams.update({'font.size': 10, 'figure.titlesize': 14, 'axes.titlesize': 12})

def plot_compact_analysis(df, features, target_col='class'):
    """
    Gera visualizações compactas adequadas para relatórios técnicos.
    Separa a análise em duas figuras: Distribuições e Normalidade (Q-Q).
    """
    n_features = len(features)
    n_cols = 3
    n_rows = math.ceil(n_features / n_cols)

    # --- FIGURA 1: Distribuições Condicionais (Histogramas/KDE) ---
    fig_hist, axes_hist = plt.subplots(n_rows, n_cols, figsize=(15, 3.5 * n_rows))
    axes_hist = axes_hist.flatten() # Facilita a iteração

    for i, col in enumerate(features):
        ax = axes_hist[i]

        # Plotando as densidades condicionais (p(x|w_i))
        sns.histplot(
            data=df, x=col, hue=target_col,
            kde=True, element="step", stat='density',
            common_norm=False, ax=ax, palette='Set1', alpha=0.3
        )
        ax.set_title(f'Dist. Condicional: {col}')
        ax.set_xlabel('')

        # Remove a legenda de todos exceto o primeiro para economizar espaço
        if i > 0 and ax.get_legend() is not None:
            ax.get_legend().remove()

    # Remove eixos vazios se houver
```

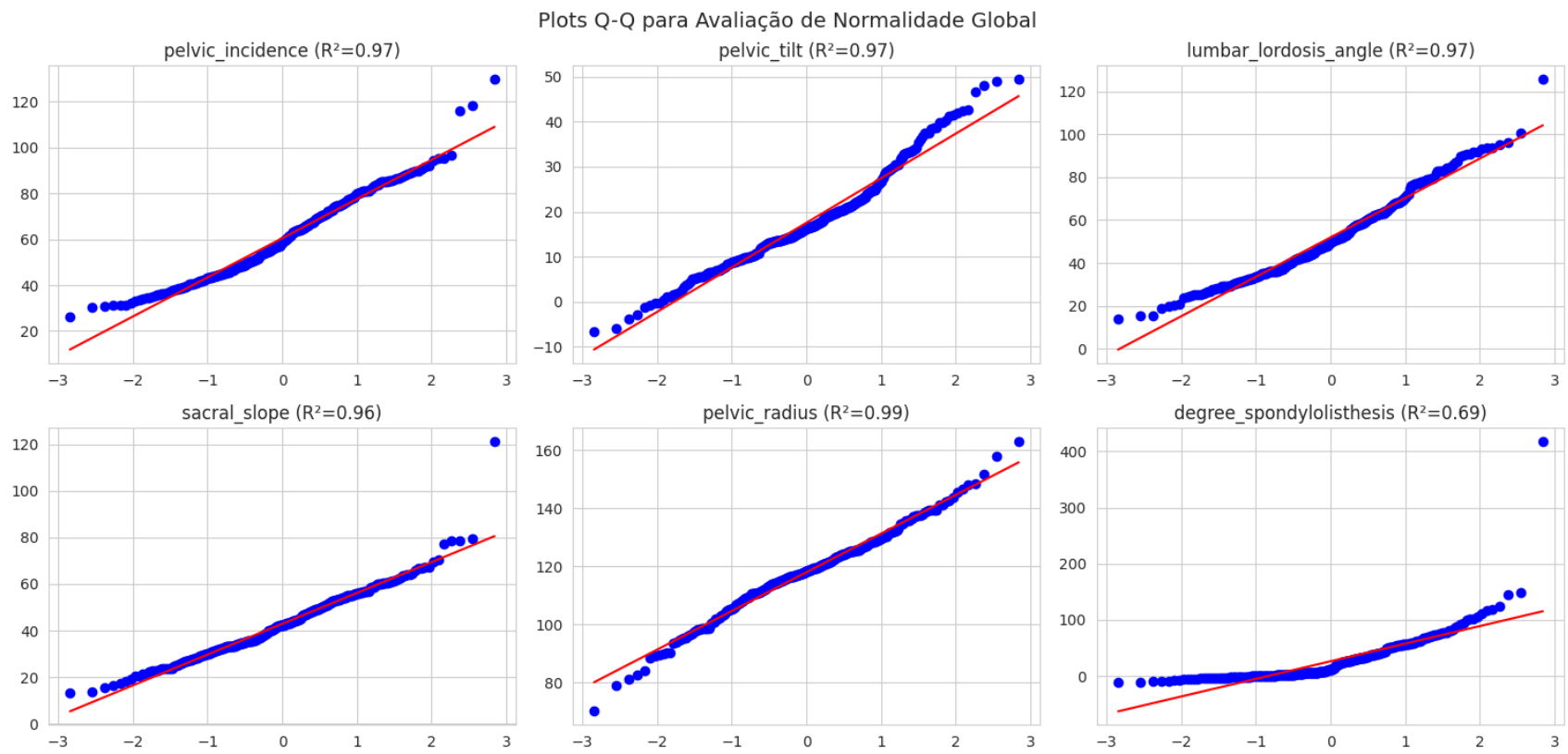
```python
    for j in range(i + 1, len(axes_hist)):
        fig_hist.delaxes(axes_hist[j])

    plt.tight_layout()
    plt.suptitle("Análise de Densidade Condicional por Classe (p(x|ω))", y=1.02)
    plt.show()

    # --- FIGURA 2: Plots Q-Q (Avaliação de Normalidade Global) ---
    fig_qq, axes_qq = plt.subplots(n_rows, n_cols, figsize=(15, 3.5 * n_rows))
    axes_qq = axes_qq.flatten()

    shapiro_results = []

    for i, col in enumerate(features):
        ax = axes_qq[i]

        # Q-Q Plot
        (osm, osr), (slope, intercept, r) = stats.probplot(df[col], dist="norm", plot=ax)
        ax.set_title(f'{col} (R²={r**2:.2f})')
        ax.set_xlabel('')
        ax.set_ylabel('')

        # Teste estatístico (apenas para registro na tabela depois)
        stat, p_value = stats.shapiro(df[col])
        shapiro_results.append({
            'Atributo': col,
            'W-Stat': f"{stat:.3f}",
            'p-value': f"{p_value:.3e}",
            'H0 (Normal)': "Não Rejeita" if p_value > 0.05 else "Rejeita"
        })

    # Remove eixos vazios
    for j in range(i + 1, len(axes_qq)):
        fig_qq.delaxes(axes_qq[j])

    plt.tight_layout()
    plt.suptitle("Plots Q-Q para Avaliação de Normalidade Global", y=1.02)
    plt.show()

    return pd.DataFrame(shapiro_results)

# --- Execução ---
 # 2. Estatísticas Descritivas

 # 1. Carregamento e Preparação dos Dados

print("--- Carregando Dataset ---")

vertebral_column = fetch_ucirepo(id=212)

df = vertebral_column.data.features.copy()
features = df.columns.tolist()

# Adicionando o target ao dataframe para facilitar visualizações

df['class'] = vertebral_column.data.targets.to_numpy().ravel()

display(df.describe().T)

print("\n=== 3. Análise Visual Compacta para Relatório ===")
df_shapiro = plot_compact_analysis(df, features, target_col='class')

print("\n=== Teste de Normalidade (Shapiro-Wilk) ===")
# Dica: Em relatórios, prefira exibir p-values muito pequenos como < 0.001
display(df_shapiro)
```

```
--- Carregando Dataset ---
```

	count	mean	std	min	25%	50%	75%	max
pelvic_incidence	310.0	60.496653	17.236520	26.147921	46.430294	58.691038	72.877696	129.834041
pelvic_tilt	310.0	17.542822	10.008330	-6.554948	10.667069	16.357689	22.120395	49.431864
lumbar_lordosis_angle	310.0	51.930930	18.554064	14.000000	37.000000	49.562398	63.000000	125.742385
sacral_slope	310.0	42.953831	13.423102	13.366931	33.347122	42.404912	52.695888	121.429566
pelvic_radius	310.0	117.920655	13.317377	70.082575	110.709196	118.268178	125.467674	163.071041
degree_spondylolisthesis	310.0	26.296694	37.559027	-11.058179	1.603727	11.767934	41.287352	418.543082

```
=== 3. Análise Visual Compacta para Relatório ===
```






```
=== Teste de Normalidade (Shapiro-Wilk) ===
```

	Atributo	W-Stat	p-value	H0 (Normal)
0	pelvic_incidence	0.971	7.132e-06	Rejeita
1	pelvic_tilt	0.966	1.321e-06	Rejeita
2	lumbar_lordosis_angle	0.972	9.221e-06	Rejeita
3	sacral_slope	0.964	5.887e-07	Rejeita
4	pelvic_radius	0.989	1.661e-02	Rejeita
5	degree_spondylolisthesis	0.697	2.908e-23	Rejeita

2. Definição e Execução dos Experimentos

In [13]:
```
def run_experiments(X, y, test_size, n_runs=100):
    le = LabelEncoder()
    y = le.fit_transform(y.ravel())

    minkowski_orders = [0.5, 2/3, 1, 3/2, 2, 5/2]
    classifiers = {f'KNN (m={m:.2f})': KNN for m in minkowski_orders}
    classifiers['MDC'] = MDC
    classifiers['MDC Robusto'] = (MDC, {'robustVersion': True})
    classifiers['MAXCO'] = MAXCO

    results = {name: [] for name in classifiers.keys()}

    for i in range(n_runs):
        X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=test_size, random_state=i, stra

        for name, clf_class in classifiers.items():
            params = {}
            if isinstance(clf_class, tuple):
                clf_class, params = clf_class

            # Treinamento
            start_train_time = time.time()
            if 'KNN' in name:
                m = float(name.split('=')[1][:-1])
```

```
            model = clf_class(X=X_train, Y=y_train, k=1, minkowskiOrder=m)
        elif 'MDC' in name:
            model = clf_class(X=X_train, Y=y_train, **params)
            model.train()
        elif 'MAXCO' in name:
            model = clf_class(X=X_train, Y=y_train)
            model.train()
        train_time = time.time() - start_train_time

        # Teste
        start_test_time = time.time()
        y_pred = model.predict(X_test)
        test_time = (time.time() - start_test_time) / len(X_test)

        # Salvar resultados
        accuracy = accuracy_score(y_test, y_pred)
        results[name].append({
            'run': i,
            'accuracy': accuracy,
            'train_time': train_time * 1000,
            'test_time': test_time * 1000,
            'y_test': y_test,
            'y_pred': y_pred
        })
    return results, le
```

3. Análise dos Resultados

In [14]:
```
def analyze_results(results, le, train_ratio):
    summary_data = []
    for name, runs in results.items():
        accuracies = [r['accuracy'] for r in runs]
        train_times = [r['train_time'] for r in runs]
        test_times = [r['test_time'] for r in runs]

        # OBS 1 & 2
        summary = {
            'Classifier': name,
            'Train Ratio': train_ratio,
            'Acuracia Media': np.mean(accuracies),
            'Std Acuracia': np.std(accuracies),
            'Min Acuracia': np.min(accuracies),
            'Max Acuracia': np.max(accuracies),
            'Median Acuracia': np.median(accuracies),
            'Tempo de Treino Medio (ms)': np.mean(train_times),
            'Tempo de Treino Std (ms)': np.std(train_times),
            'Tempo de Teste Medio (ms)': np.mean(test_times),
            'Tempo de Teste Std (ms)': np.std(test_times),
        }
        summary_data.append(summary)

    summary_df = pd.DataFrame(summary_data).sort_values(by='Acuracia Media', ascending=False)
    display(summary_df.drop(columns=['Train Ratio']))

    # OBS 2: Matrizes de confusão para melhor e pior rodada
    for name, runs in results.items():
        accuracies = [r['accuracy'] for r in runs]
        best_run_idx = np.argmax(accuracies)
        worst_run_idx = np.argmin(accuracies)
        best_run = runs[best_run_idx]
        worst_run = runs[worst_run_idx]

        print(f'--- {name} ---')
        print(f'Melhor rodada (#{best_run["run"]}) Acurácia: {best_run["accuracy"]:.4f}')
        print(f"Pior rodada (#{worst_run['run']}) Acurácia: {worst_run['accuracy']:.4f}")

        fig, axes = plt.subplots(1, 2, figsize=(15, 5))
        cm_best = confusion_matrix(best_run['y_test'], best_run['y_pred'])
        sns.heatmap(cm_best, annot=True, fmt='d', cmap='Blues', xticklabels=le.classes_, yticklabels=le.cla
        axes[0].set_title(f'Matriz de Confusão para a melhor rodada')
        axes[0].set_xlabel('Predito')
        axes[0].set_ylabel('Verdadeiro')

        cm_worst = confusion_matrix(worst_run['y_test'], worst_run['y_pred'])
        sns.heatmap(cm_worst, annot=True, fmt='d', cmap='Reds', xticklabels=le.classes_, yticklabels=le.cla
        axes[1].set_title(f'Matriz de Confusão para a pior rodada')
        axes[1].set_xlabel('Predito')
        axes[1].set_ylabel('Verdadeiro')
        plt.tight_layout()
        plt.show()

        # OBS 4: Estatísticas por classe (para a melhor rodada)
        print(f'Classification Report for Best Run (#{best_run["run"]}):')
        print(classification_report(best_run['y_test'], best_run['y_pred'], target_names=le.classes_))

    return summary_df
```

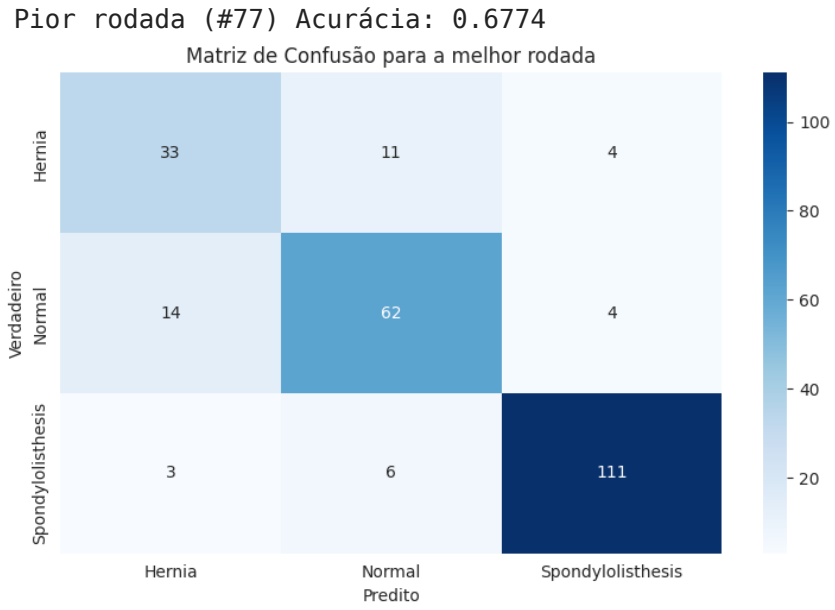
4. Execução Principal e Análise por Split de Teste

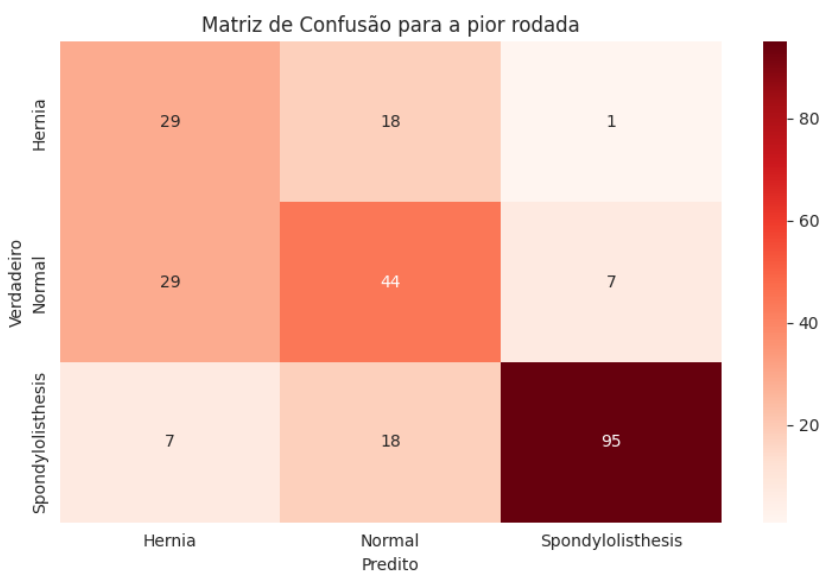
In [15]:
```
test_splits = [0.8, 0.7, 0.5, 0.3, 0.2] # Corresponde a 20/80, 30/70, 50/50, 70/30, 80/20 train/test
all_split_results = {}

resultados_df = pd.DataFrame()

for split in test_splits:
    train_ratio = int(split*100)
    test_ratio = int((1-split)*100)
    print(f'====================================================================')
    print(f'   Rodando Experimento para Divisão: {train_ratio}/{test_ratio}   ')
    print(f'====================================================================')

    results, le = run_experiments(X, y_raw, test_size=split, n_runs=100)
    resultado = analyze_results(results, le, train_ratio)

    resultados_df = pd.concat([resultados_df, resultado], ignore_index=True)
    all_split_results[f'{train_ratio}/{test_ratio}'] = results
```

```
====================================================================
   Rodando Experimento para Divisão: 80/19
====================================================================
```

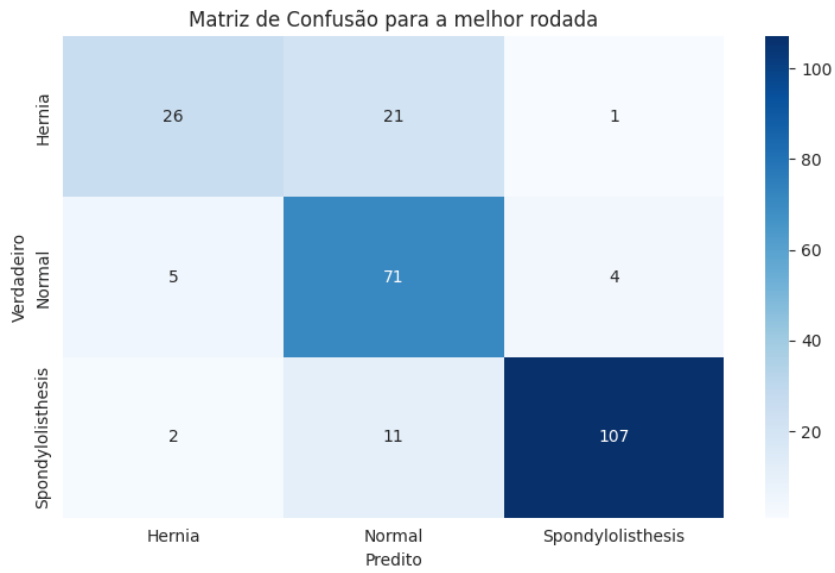
	Classifier	Acuracia Media	Std Acuracia	Min Acuracia	Max Acuracia	Median Acuracia	Tempo de Treino Medio (ms)	Tempo de Treino Std (ms)	Tempo de Teste Medio (ms)	Tempo de Teste Std (ms)
4	KNN (m=2.00)	0.784516	0.028124	0.685484	0.834677	0.786290	0.002663	0.000440	0.019282	0.000380
3	KNN (m=1.50)	0.784516	0.029709	0.669355	0.842742	0.790323	0.002599	0.000444	0.024999	0.001487
5	KNN (m=2.50)	0.783669	0.027309	0.673387	0.834677	0.786290	0.002627	0.000441	0.024817	0.000601
2	KNN (m=1.00)	0.778710	0.029134	0.693548	0.846774	0.778226	0.002716	0.000545	0.019745	0.001175
7	MDC Robusto	0.772944	0.033970	0.673387	0.846774	0.776210	0.122716	0.007636	0.005350	0.000045
1	KNN (m=0.67)	0.765847	0.027213	0.689516	0.822581	0.768145	0.002780	0.000535	0.025164	0.001632
6	MDC	0.760565	0.045114	0.592742	0.846774	0.764113	0.062957	0.004925	0.005377	0.000112
0	KNN (m=0.50)	0.756815	0.028247	0.677419	0.830645	0.758065	0.003786	0.000894	0.019871	0.000788
8	MAXCO	0.713024	0.052021	0.540323	0.802419	0.719758	0.049629	0.002506	0.036097	0.000444

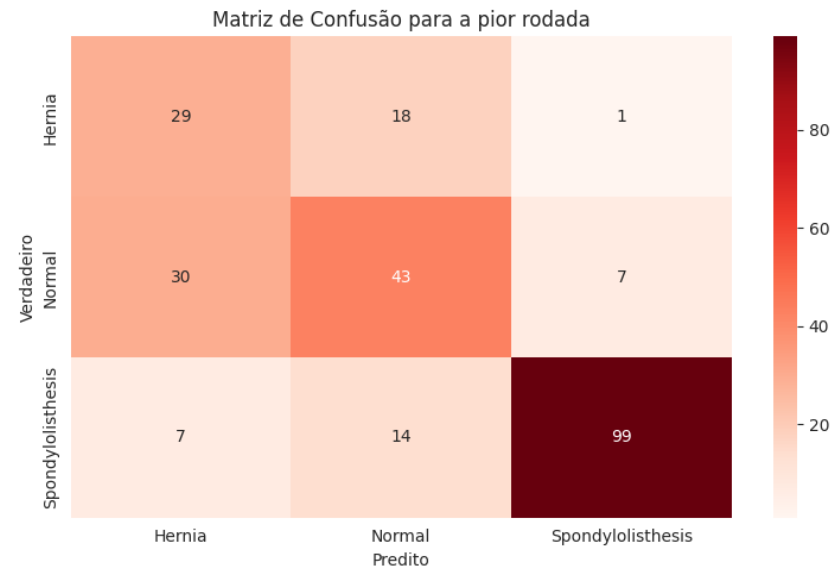
```
--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8306
Pior rodada (#77) Acurácia: 0.6774
```



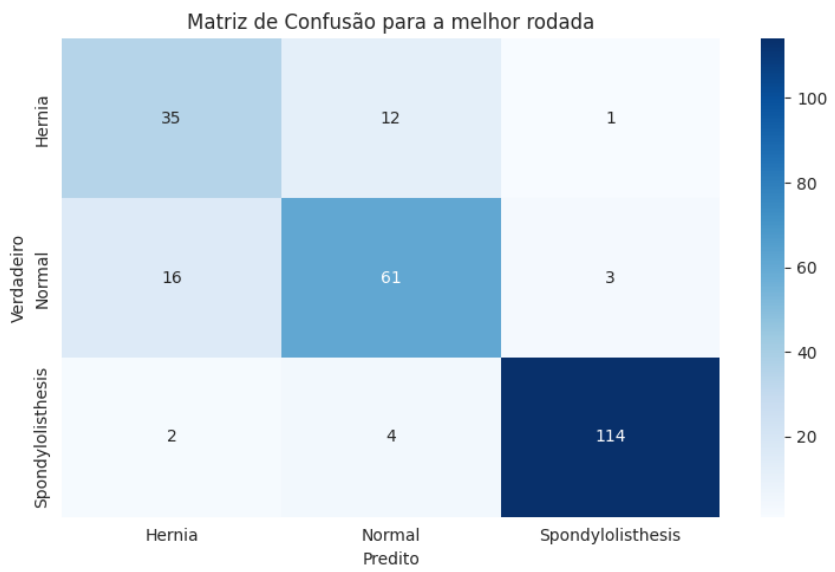


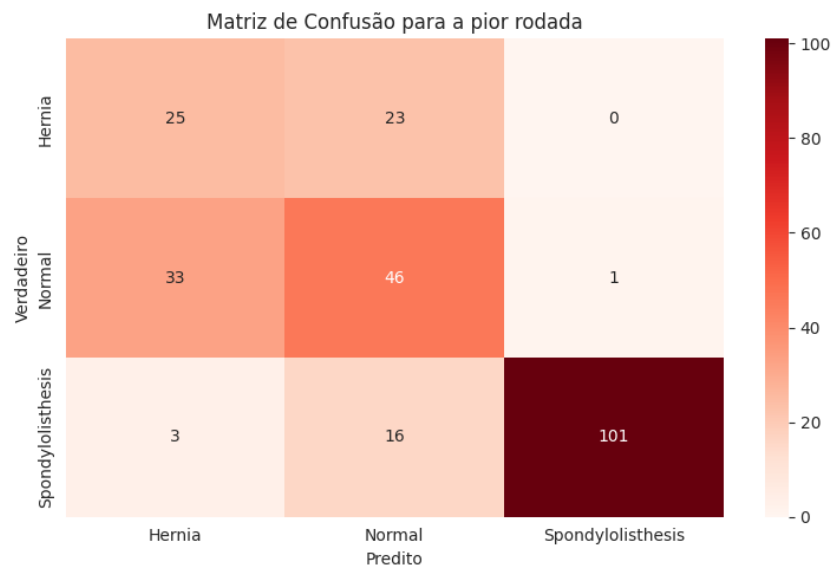

```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.66      0.69      0.67        48
           Normal       0.78      0.78      0.78        80
Spondylolisthesis       0.93      0.93      0.93       120

         accuracy                           0.83       248
        macro avg       0.79      0.80      0.79       248
     weighted avg       0.83      0.83      0.83       248

--- KNN (m=0.67) ---
Melhor rodada (#0) Acurácia: 0.8226
Pior rodada (#77) Acurácia: 0.6895
```



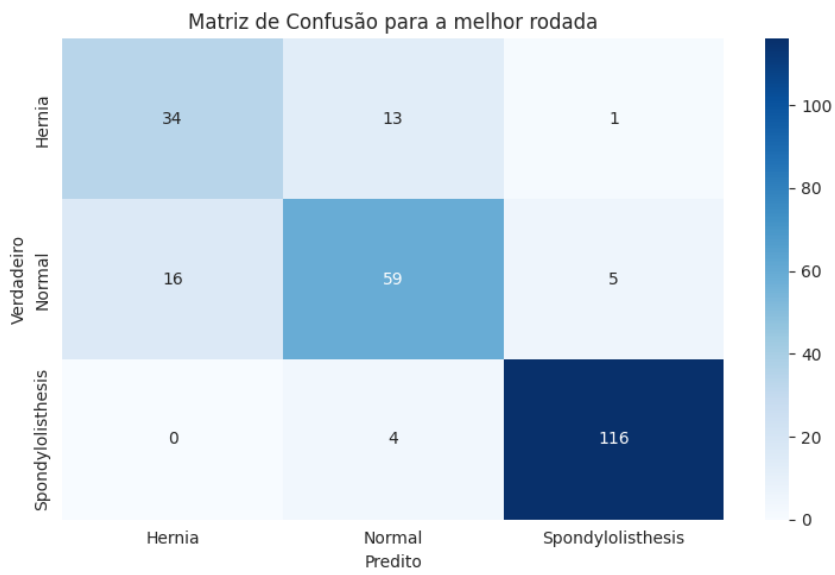


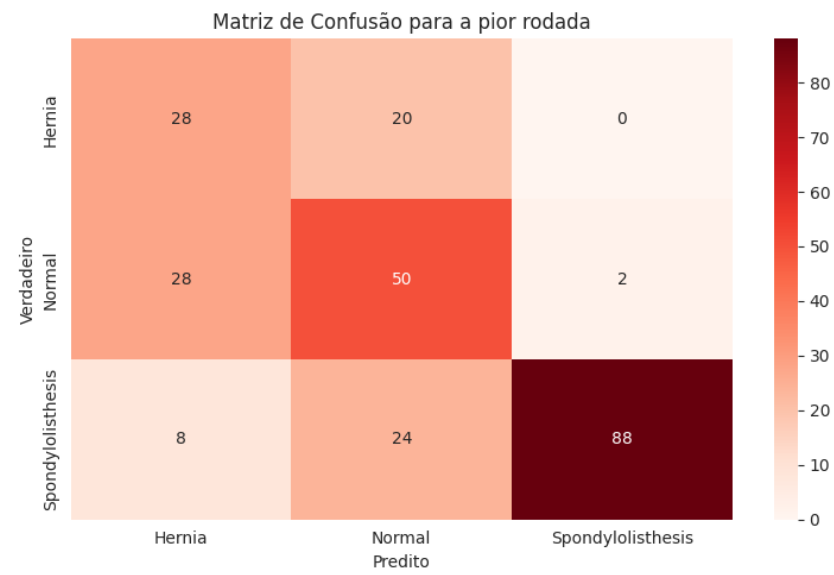

```
Classification Report for Best Run (#0):
                   precision    recall  f1-score   support

           Hernia       0.79      0.54      0.64        48
           Normal       0.69      0.89      0.78        80
Spondylolisthesis       0.96      0.89      0.92       120

         accuracy                           0.82       248
        macro avg       0.81      0.77      0.78       248
     weighted avg       0.84      0.82      0.82       248

--- KNN (m=1.00) ---
Melhor rodada (#98) Acurácia: 0.8468
Pior rodada (#22) Acurácia: 0.6935
```



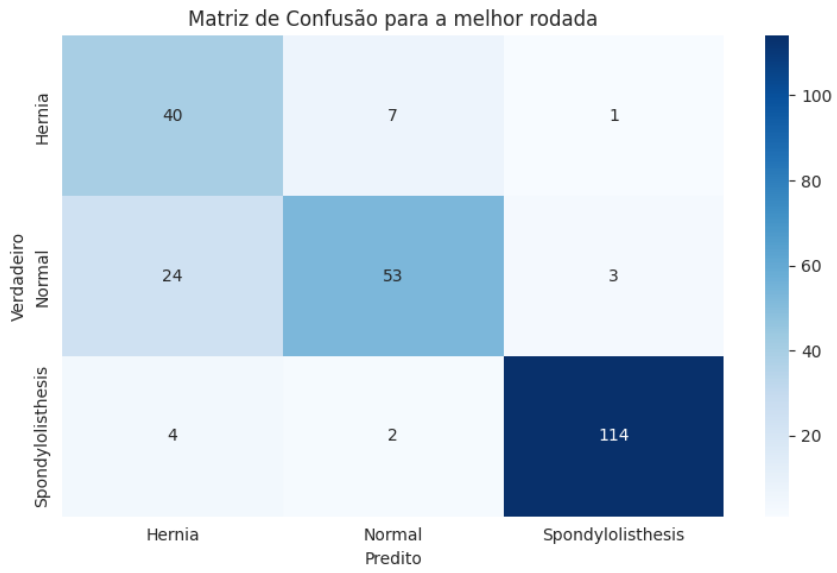


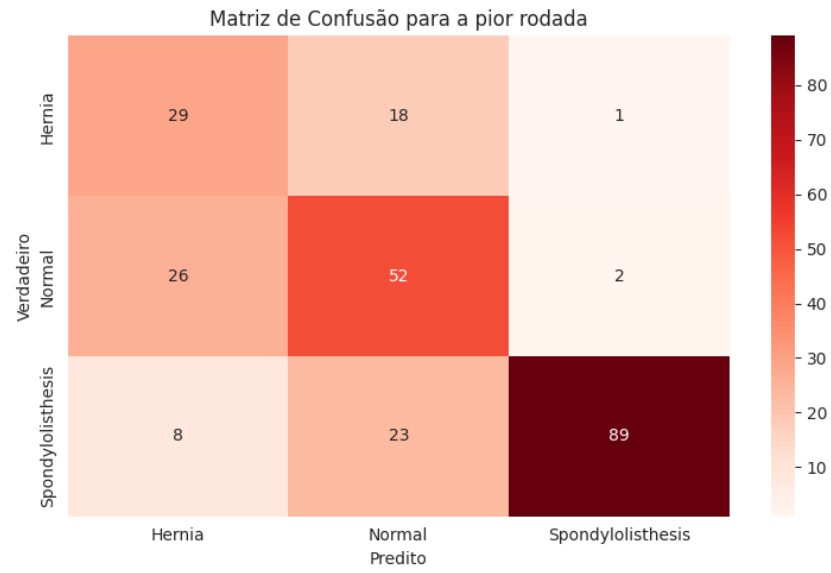

```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.66      0.73      0.69        48
           Normal       0.79      0.76      0.78        80
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.85       248
        macro avg       0.81      0.81      0.81       248
     weighted avg       0.85      0.85      0.85       248

--- KNN (m=1.50) ---
Melhor rodada (#1) Acurácia: 0.8427
Pior rodada (#31) Acurácia: 0.6694
```



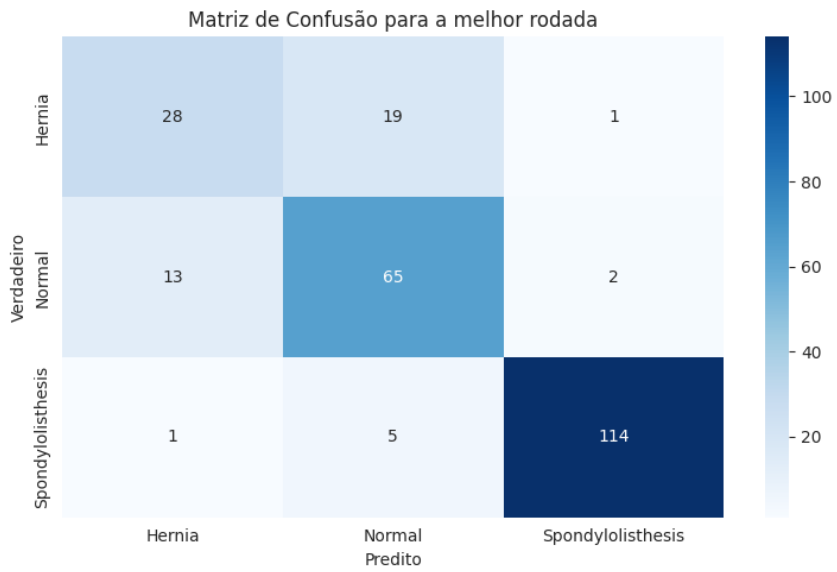


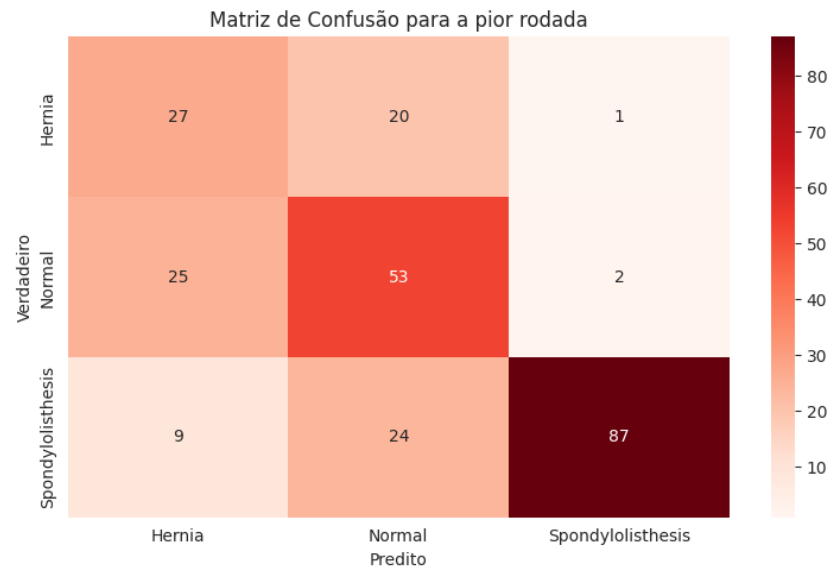
```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.68      0.71      0.69        48
           Normal       0.78      0.74      0.76        80
Spondylolisthesis       0.95      0.97      0.96       120

         accuracy                           0.84       248
        macro avg       0.80      0.80      0.80       248
     weighted avg       0.84      0.84      0.84       248

--- KNN (m=2.00) ---
Melhor rodada (#87) Acurácia: 0.8347
Pior rodada (#31) Acurácia: 0.6855
```



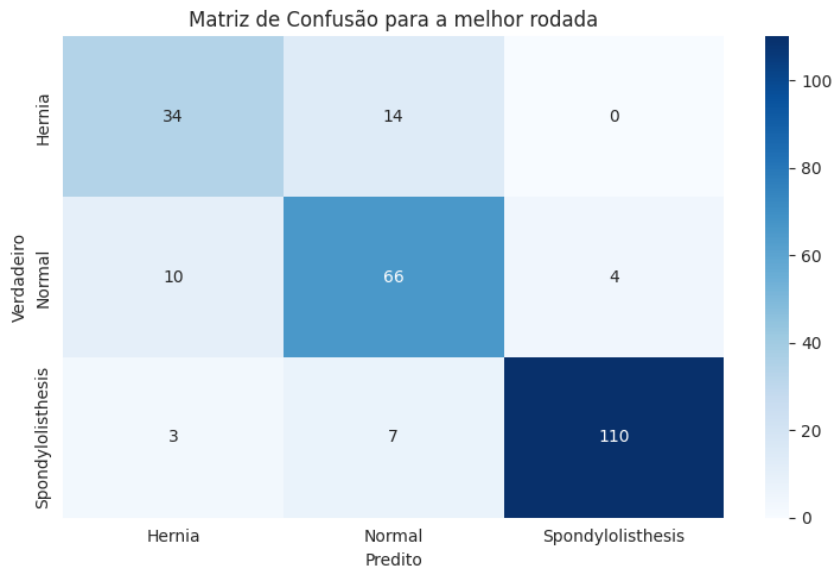


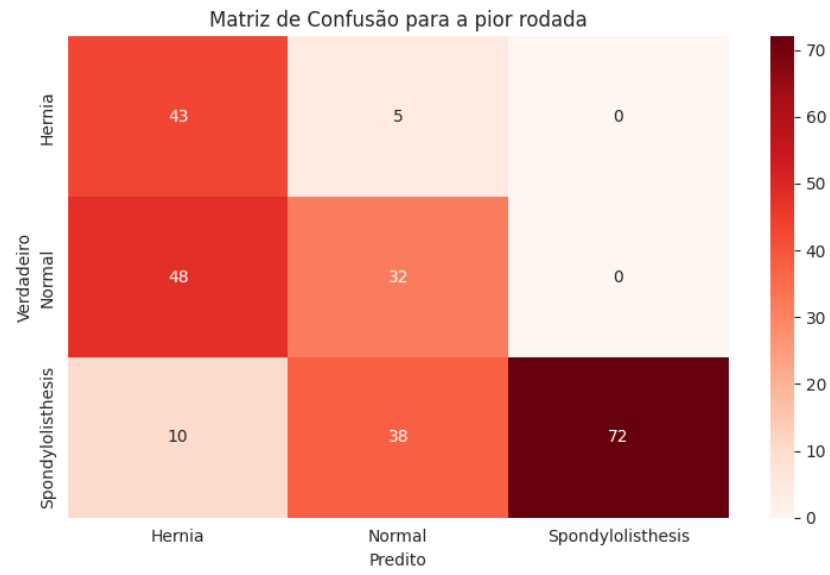

```
Classification Report for Best Run (#87):
                   precision    recall  f1-score   support

           Hernia       0.59      0.83      0.69        48
           Normal       0.85      0.66      0.75        80
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.83       248
        macro avg       0.80      0.82      0.80       248
     weighted avg       0.86      0.83      0.84       248

--- KNN (m=2.50) ---
Melhor rodada (#6) Acurácia: 0.8347
Pior rodada (#31) Acurácia: 0.6734
```



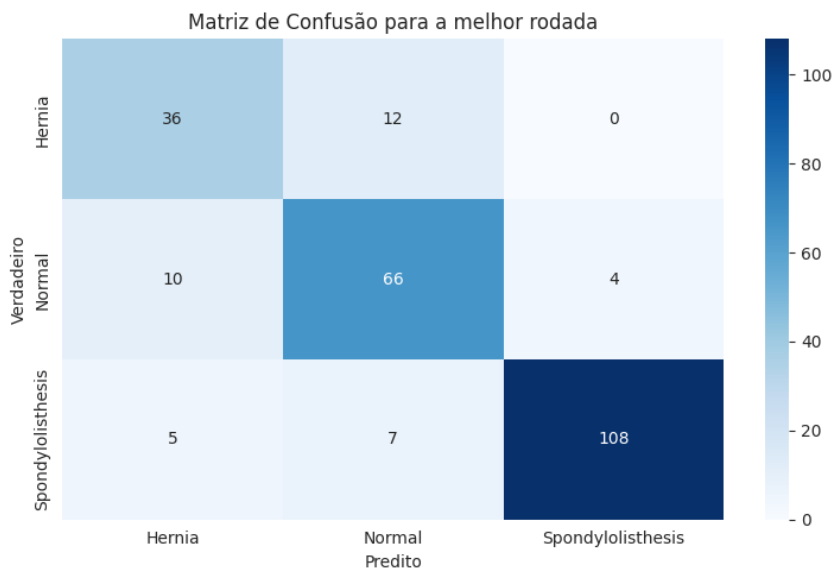


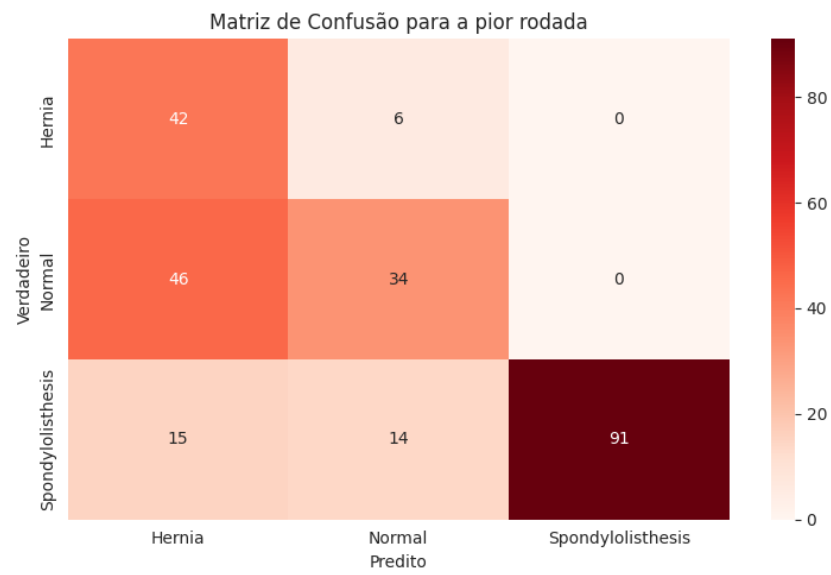

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.67      0.58      0.62        48
           Normal       0.73      0.81      0.77        80
Spondylolisthesis       0.97      0.95      0.96       120

         accuracy                           0.83       248
        macro avg       0.79      0.78      0.78       248
     weighted avg       0.84      0.83      0.83       248

--- MDC ---
Melhor rodada (#6) Acurácia: 0.8468
Pior rodada (#31) Acurácia: 0.5927
```



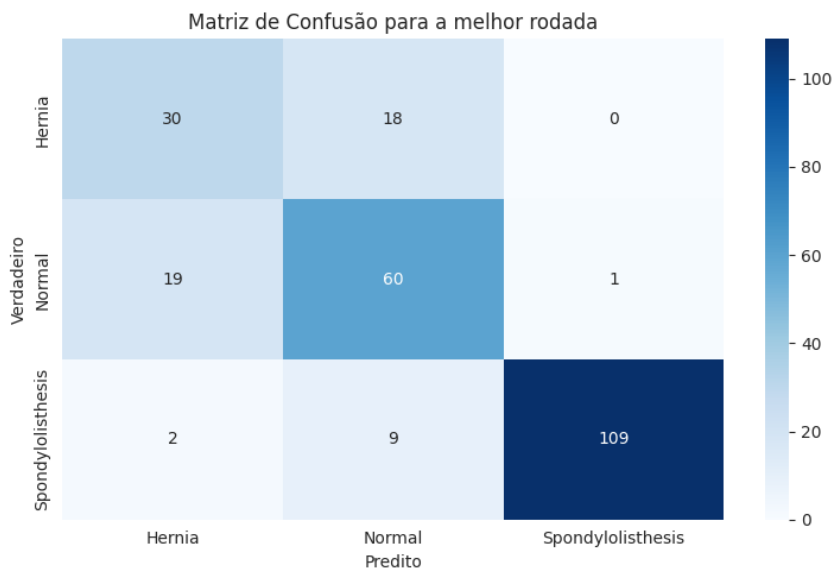


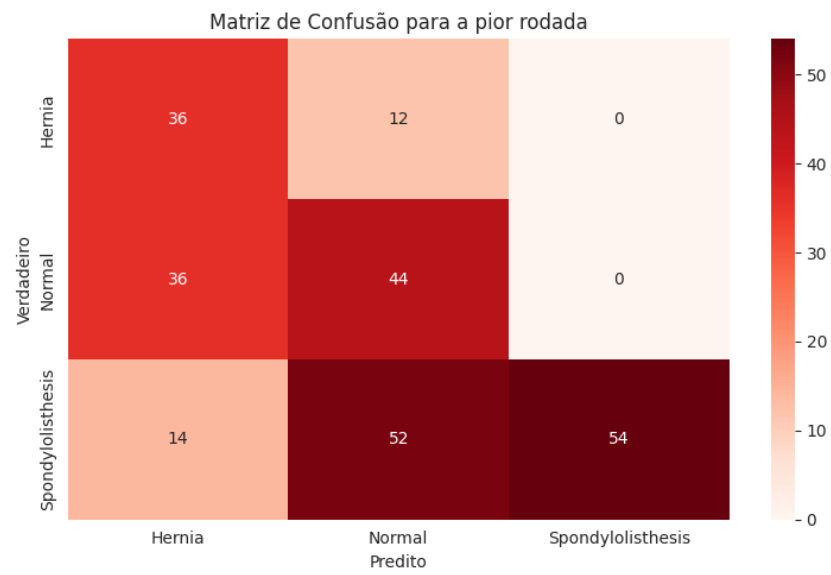

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.72      0.71      0.72        48
           Normal       0.76      0.82      0.79        80
Spondylolisthesis       0.96      0.92      0.94       120

         accuracy                           0.85       248
        macro avg       0.82      0.82      0.82       248
     weighted avg       0.85      0.85      0.85       248

--- MDC Robusto ---
Melhor rodada (#83) Acurácia: 0.8468
Pior rodada (#31) Acurácia: 0.6734
```






```
Classification Report for Best Run (#83):
                   precision    recall  f1-score   support

           Hernia       0.71      0.75      0.73        48
           Normal       0.78      0.82      0.80        80
Spondylolisthesis       0.96      0.90      0.93       120

         accuracy                           0.85       248
        macro avg       0.82      0.83      0.82       248
     weighted avg       0.85      0.85      0.85       248

--- MAXCO ---
Melhor rodada (#84) Acurácia: 0.8024
Pior rodada (#76) Acurácia: 0.5403
```

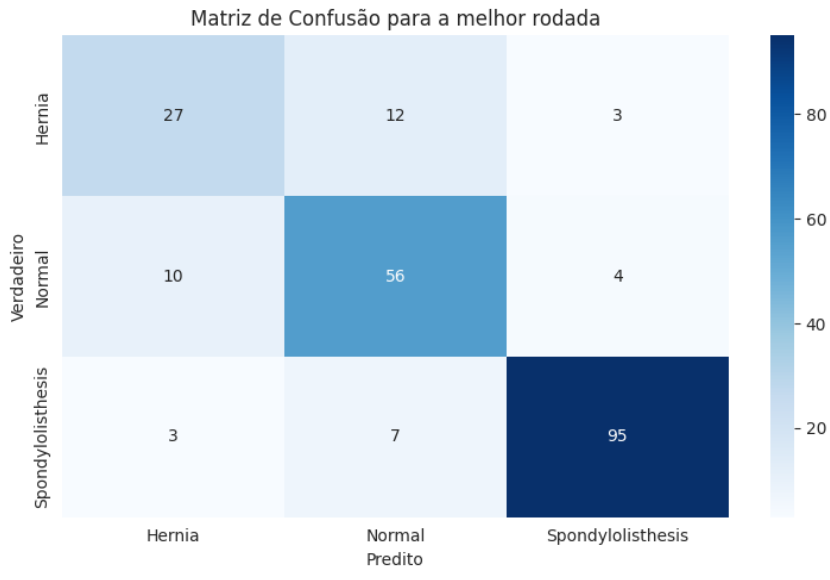


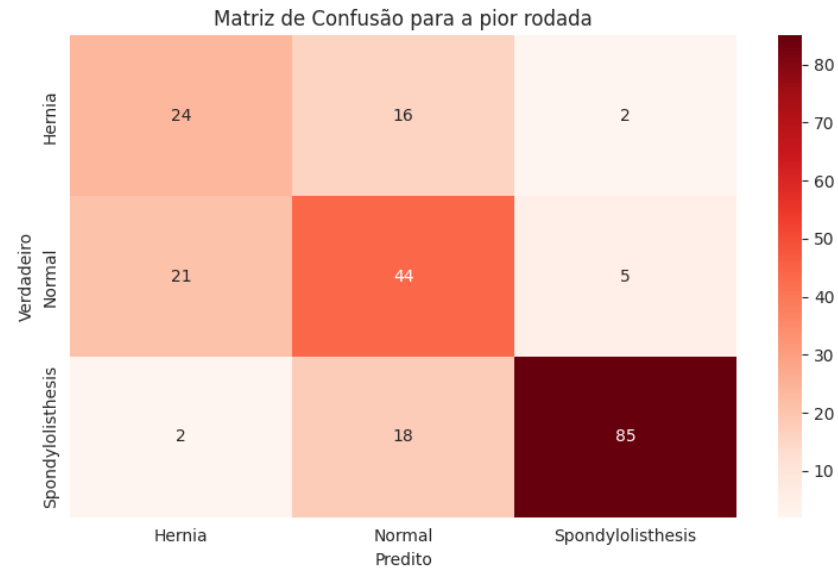


```
Classification Report for Best Run (#84):
                   precision    recall  f1-score   support

           Hernia       0.59      0.62      0.61        48
           Normal       0.69      0.75      0.72        80
Spondylolisthesis       0.99      0.91      0.95       120

         accuracy                           0.80       248
        macro avg       0.76      0.76      0.76       248
     weighted avg       0.82      0.80      0.81       248

==================================================================
   Rodando Experimento para Divisão: 70/30
==================================================================
```

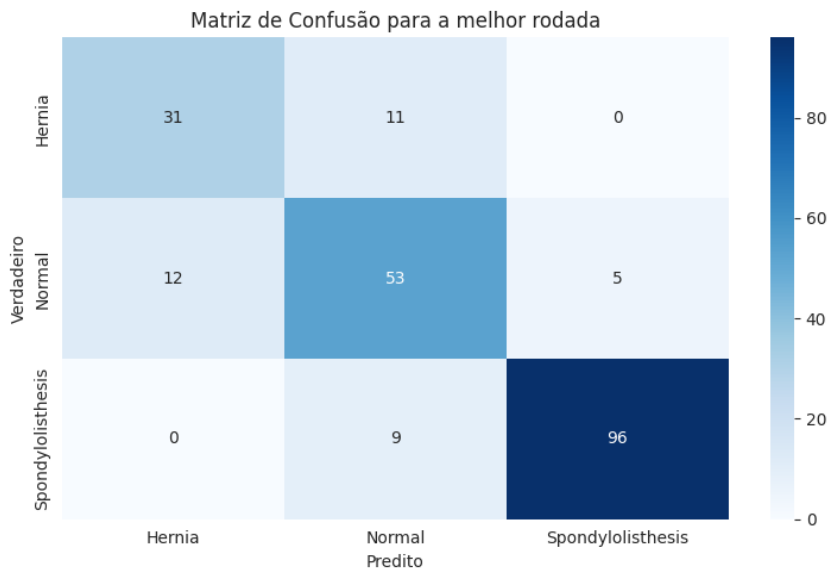
	Classifier	Acuracia Media	Std Acuracia	Min Acuracia	Max Acuracia	Median Acuracia	Tempo de Treino Medio (ms)	Tempo de Treino Std (ms)	Tempo de Teste Medio (ms)	Tempo de Teste Std (ms)
4	KNN (m=2.00)	0.794931	0.028351	0.714286	0.857143	0.797235	0.002558	0.000461	0.021382	0.000193
5	KNN (m=2.50)	0.793641	0.027523	0.695853	0.852535	0.797235	0.002451	0.000305	0.029387	0.000523
3	KNN (m=1.50)	0.792488	0.030010	0.691244	0.861751	0.792627	0.002525	0.000424	0.029365	0.000411
2	KNN (m=1.00)	0.786498	0.027811	0.700461	0.847926	0.790323	0.002542	0.000222	0.021773	0.000742
7	MDC Robusto	0.777327	0.029700	0.649770	0.838710	0.778802	0.124888	0.005391	0.005446	0.000060
1	KNN (m=0.67)	0.771935	0.024012	0.691244	0.829493	0.774194	0.002639	0.000248	0.029436	0.000172
6	MDC	0.765115	0.034837	0.663594	0.843318	0.769585	0.063791	0.006221	0.005452	0.000049
0	KNN (m=0.50)	0.760276	0.024393	0.705069	0.820276	0.764977	0.003555	0.000370	0.021965	0.000167
8	MAXCO	0.719263	0.043843	0.599078	0.829493	0.728111	0.051167	0.002975	0.036272	0.000226

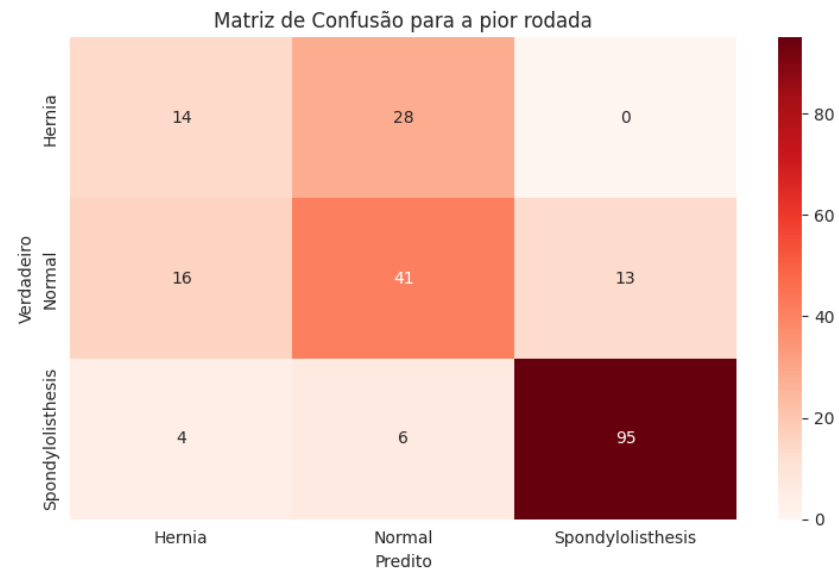
```
--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8203
Pior rodada (#34) Acurácia: 0.7051
```



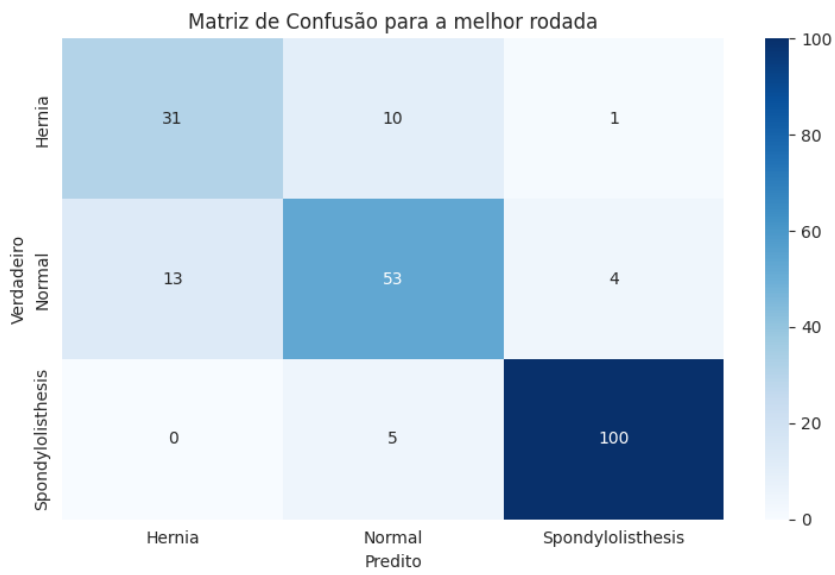


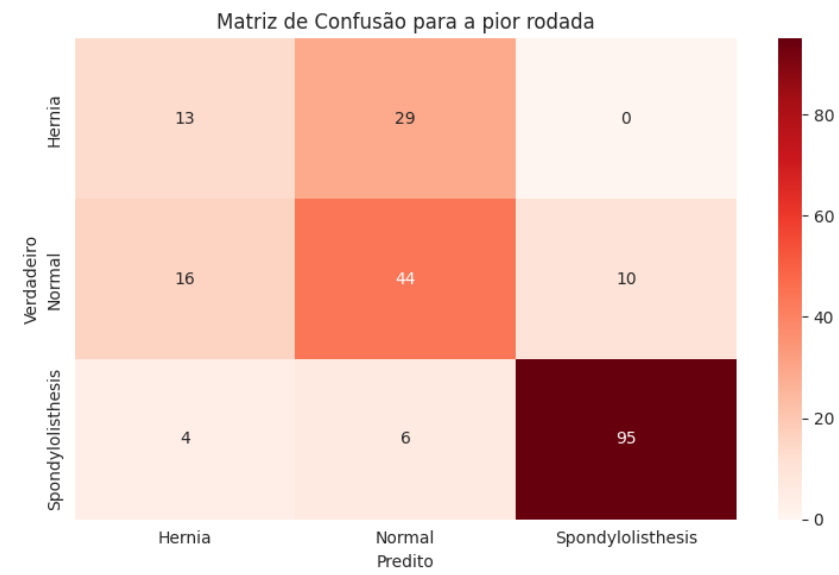

```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.68      0.64      0.66        42
           Normal       0.75      0.80      0.77        70
Spondylolisthesis       0.93      0.90      0.92       105

         accuracy                           0.82       217
        macro avg       0.78      0.78      0.78       217
     weighted avg       0.82      0.82      0.82       217

--- KNN (m=0.67) ---
Melhor rodada (#65) Acurácia: 0.8295
Pior rodada (#49) Acurácia: 0.6912
```



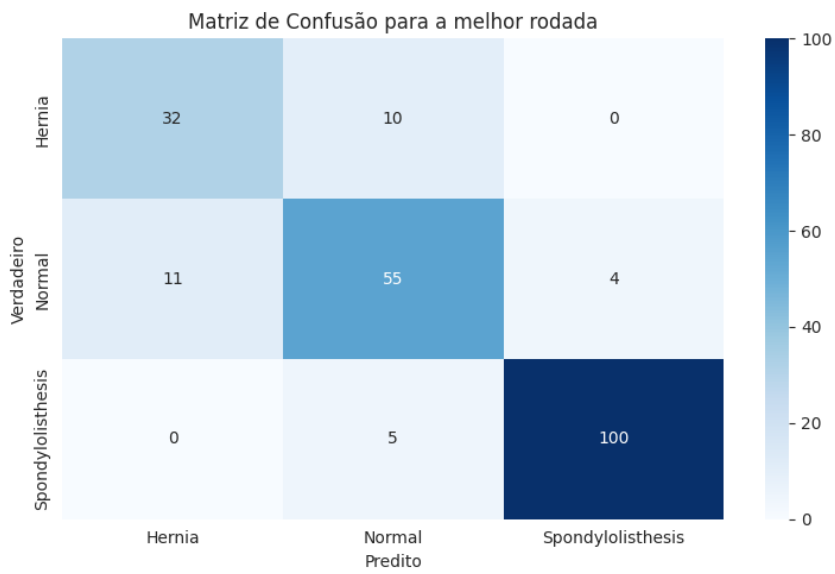


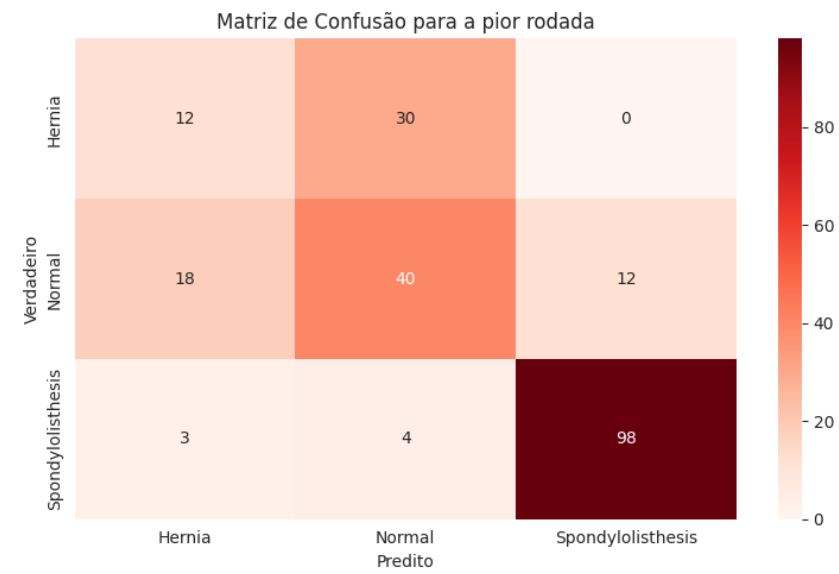

```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.72      0.74      0.73        42
           Normal       0.73      0.76      0.74        70
Spondylolisthesis       0.95      0.91      0.93       105

         accuracy                           0.83       217
        macro avg       0.80      0.80      0.80       217
     weighted avg       0.83      0.83      0.83       217

--- KNN (m=1.00) ---
Melhor rodada (#1) Acurácia: 0.8479
Pior rodada (#49) Acurácia: 0.7005
```



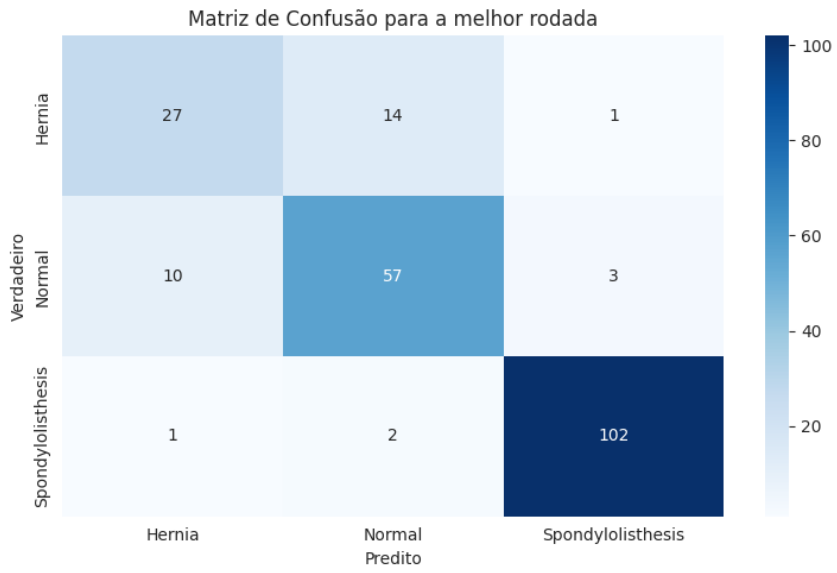


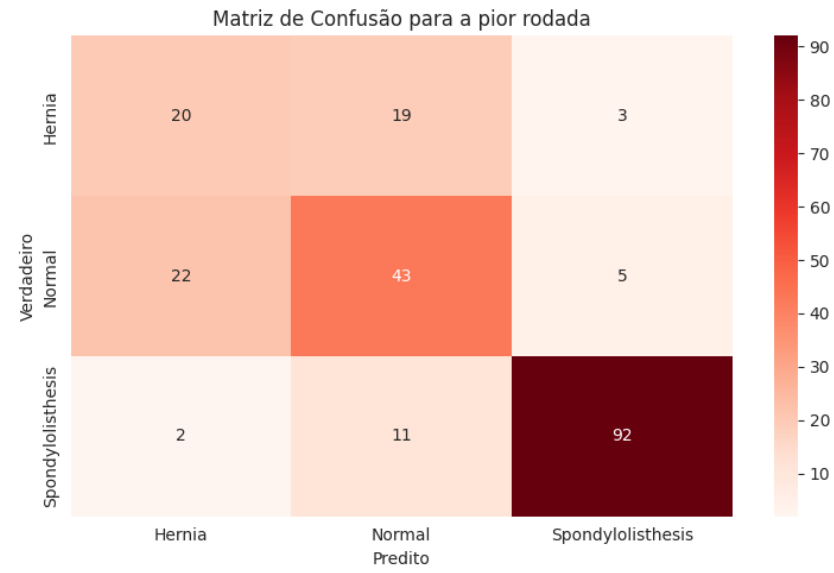

```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.70      0.74      0.72        42
           Normal       0.78      0.76      0.77        70
Spondylolisthesis       0.95      0.95      0.95       105

         accuracy                           0.85       217
        macro avg       0.81      0.82      0.81       217
     weighted avg       0.85      0.85      0.85       217

--- KNN (m=1.50) ---
Melhor rodada (#1) Acurácia: 0.8618
Pior rodada (#49) Acurácia: 0.6912
```



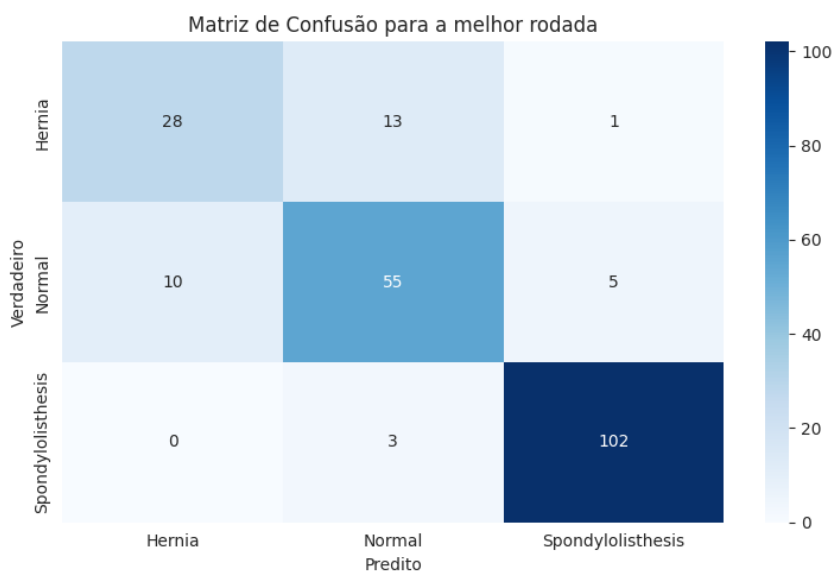


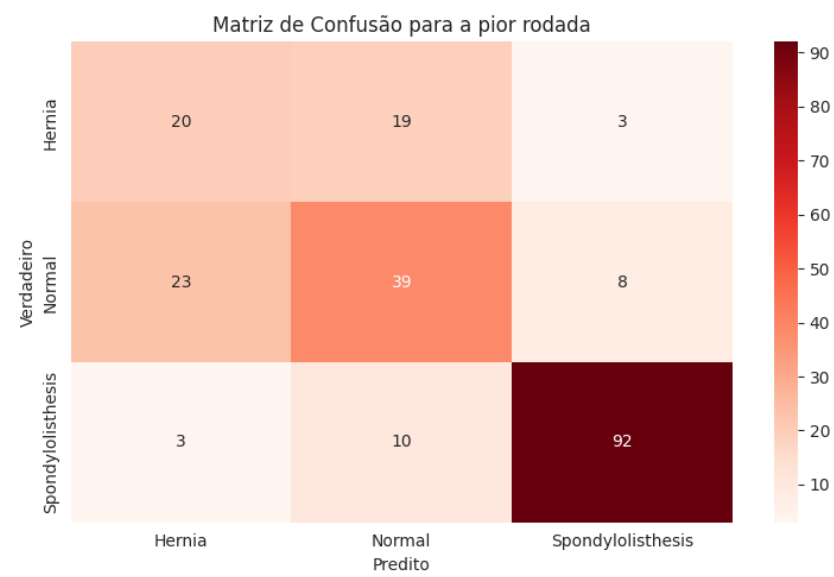
```
Classification Report for Best Run (#1):
                   precision    recall  f1-score   support

           Hernia       0.74      0.76      0.75        42
           Normal       0.79      0.79      0.79        70
Spondylolisthesis       0.96      0.95      0.96       105

         accuracy                           0.86       217
        macro avg       0.83      0.83      0.83       217
     weighted avg       0.86      0.86      0.86       217

--- KNN (m=2.00) ---
Melhor rodada (#6) Acurácia: 0.8571
Pior rodada (#42) Acurácia: 0.7143
```



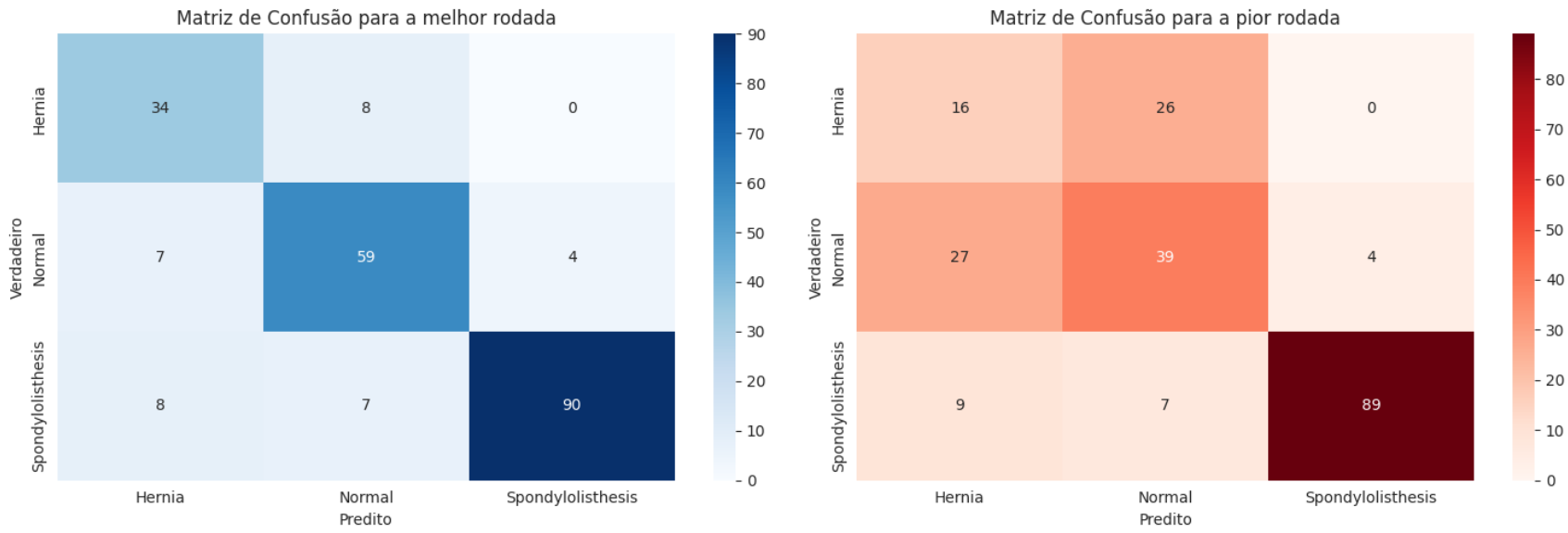



```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.71      0.64      0.68        42
           Normal       0.78      0.81      0.80        70
Spondylolisthesis       0.96      0.97      0.97       105

         accuracy                           0.86       217
        macro avg       0.82      0.81      0.81       217
     weighted avg       0.86      0.86      0.86       217

--- KNN (m=2.50) ---
Melhor rodada (#8) Acurácia: 0.8525
Pior rodada (#42) Acurácia: 0.6959
```

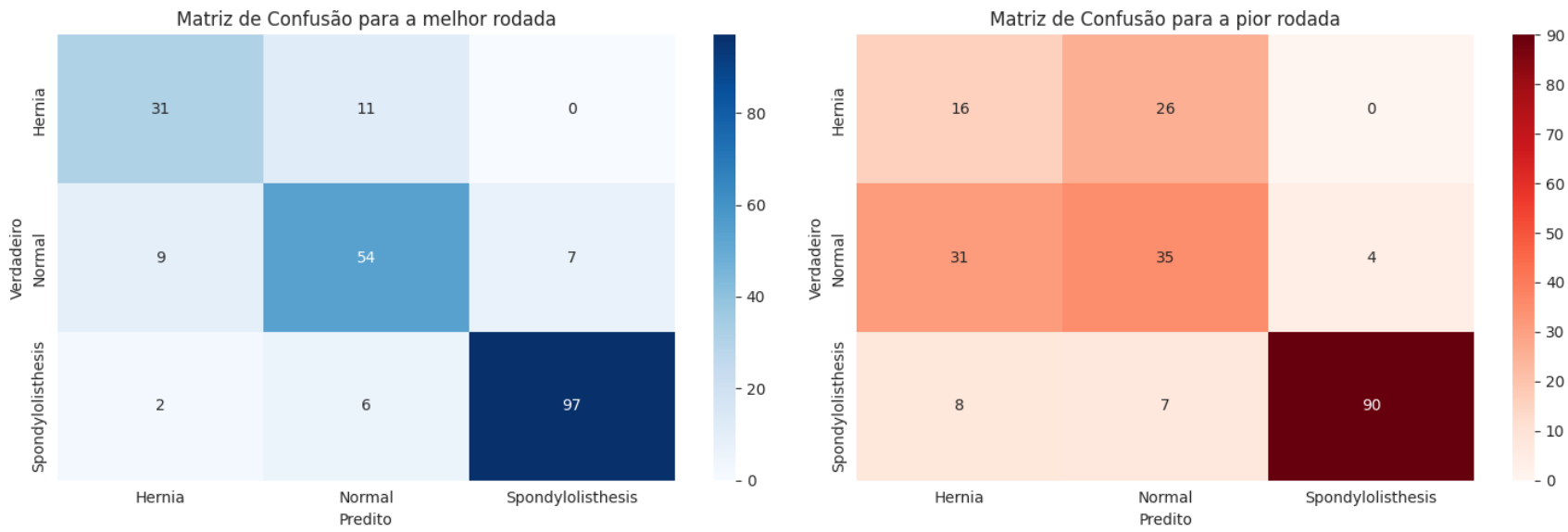




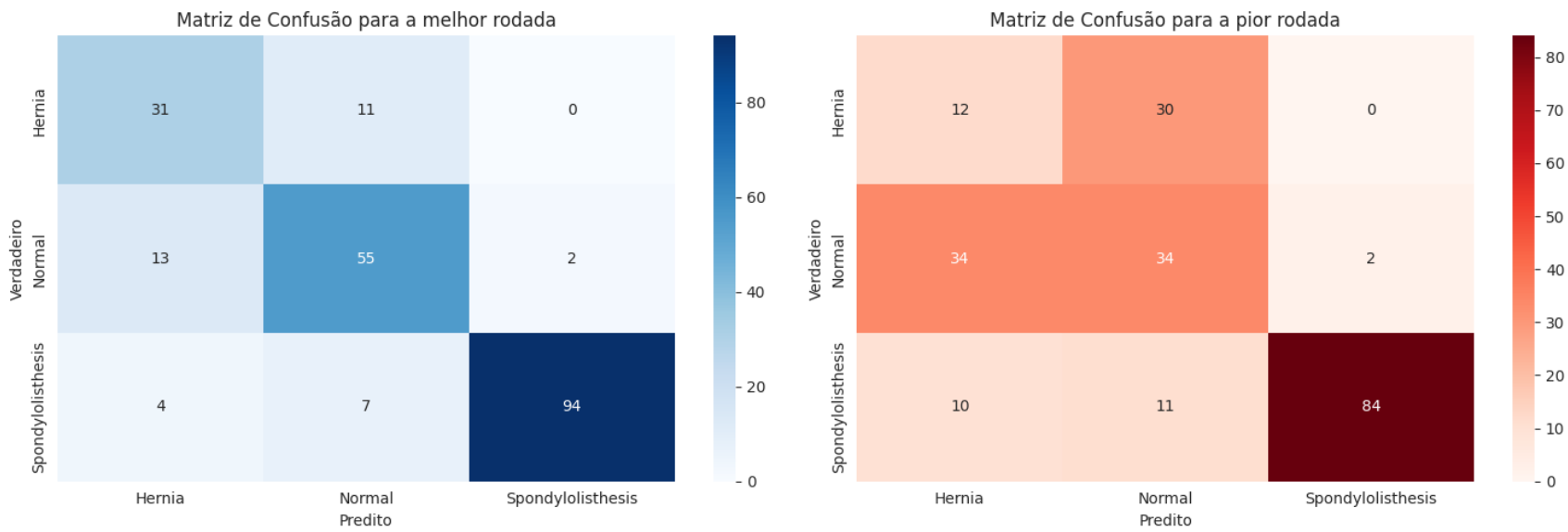

```
Classification Report for Best Run (#8):
                   precision    recall  f1-score   support

           Hernia       0.74      0.67      0.70        42
           Normal       0.77      0.79      0.78        70
Spondylolisthesis       0.94      0.97      0.96       105

         accuracy                           0.85       217
        macro avg       0.82      0.81      0.81       217
     weighted avg       0.85      0.85      0.85       217

--- MDC ---
Melhor rodada (#59) Acurácia: 0.8433
Pior rodada (#49) Acurácia: 0.6636
```




```
Classification Report for Best Run (#59):
                   precision    recall  f1-score   support

           Hernia       0.69      0.81      0.75        42
           Normal       0.80      0.84      0.82        70
Spondylolisthesis       0.96      0.86      0.90       105

         accuracy                           0.84       217
        macro avg       0.82      0.84      0.82       217
     weighted avg       0.85      0.84      0.85       217

--- MDC Robusto ---
Melhor rodada (#6) Acurácia: 0.8387
Pior rodada (#49) Acurácia: 0.6498
```



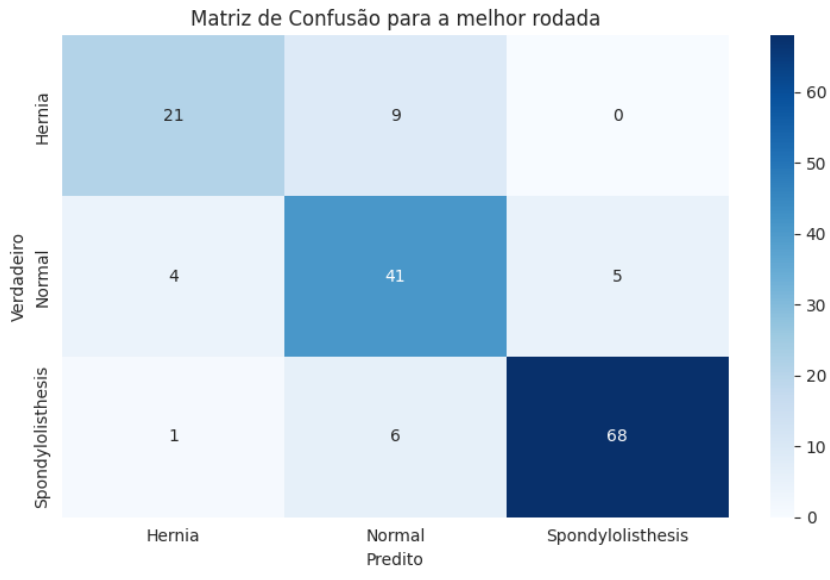

```
Classification Report for Best Run (#6):
                   precision    recall  f1-score   support

           Hernia       0.74      0.74      0.74        42
           Normal       0.76      0.77      0.77        70
Spondylolisthesis       0.93      0.92      0.93       105

         accuracy                           0.84       217
        macro avg       0.81      0.81      0.81       217
     weighted avg       0.84      0.84      0.84       217

--- MAXCO ---
Melhor rodada (#69) Acurácia: 0.8295
Pior rodada (#49) Acurácia: 0.5991
```

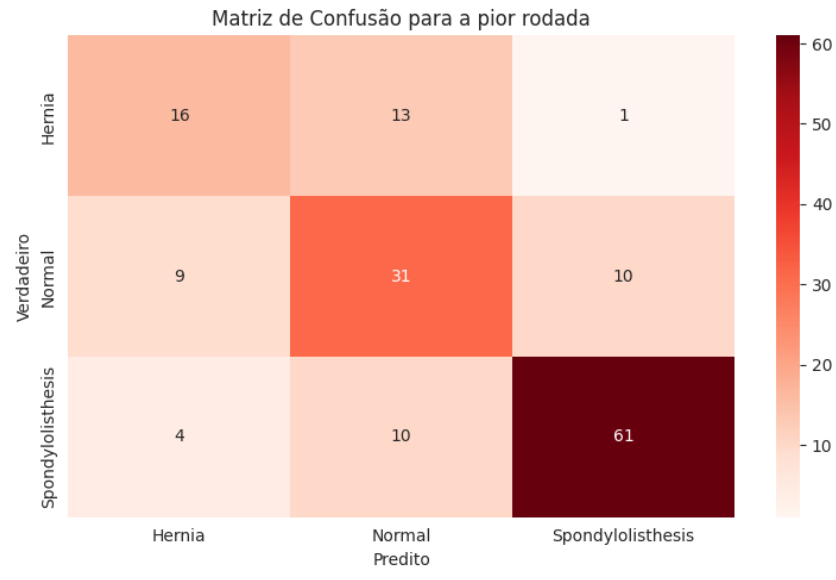


```
Classification Report for Best Run (#69):
                   precision    recall  f1-score   support

           Hernia       0.65      0.74      0.69        42
           Normal       0.75      0.79      0.77        70
Spondylolisthesis       0.98      0.90      0.94       105

         accuracy                           0.83       217
        macro avg       0.79      0.81      0.80       217
     weighted avg       0.84      0.83      0.83       217

====================================================================
   Rodando Experimento para Divisão: 50/50
====================================================================
```

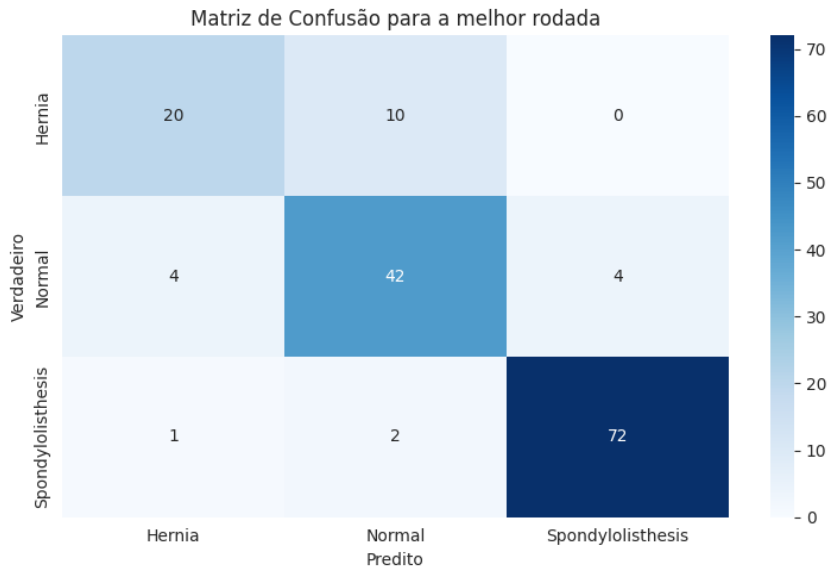
	Classifier	Acuracia Media	Std Acuracia	Min Acuracia	Max Acuracia	Median Acuracia	Tempo de Treino Medio (ms)	Tempo de Treino Std (ms)	Tempo de Teste Medio (ms)	Tempo de Teste Std (ms)
4	KNN (m=2.00)	0.806581	0.027265	0.741935	0.890323	0.806452	0.002549	0.000451	0.025654	0.000578
5	KNN (m=2.50)	0.806194	0.026880	0.741935	0.883871	0.806452	0.002532	0.000356	0.038477	0.001151
3	KNN (m=1.50)	0.804065	0.027834	0.754839	0.890323	0.806452	0.002539	0.000652	0.038489	0.001271
2	KNN (m=1.00)	0.797677	0.029403	0.722581	0.877419	0.796774	0.002632	0.000363	0.025804	0.000496
7	MDC Robusto	0.785548	0.028840	0.696774	0.851613	0.787097	0.129714	0.004919	0.005463	0.000057
1	KNN (m=0.67)	0.779613	0.027169	0.703226	0.864516	0.780645	0.002668	0.000263	0.038400	0.000490
6	MDC	0.770645	0.033751	0.677419	0.851613	0.774194	0.065167	0.005092	0.005471	0.000057
0	KNN (m=0.50)	0.766774	0.028098	0.696774	0.838710	0.767742	0.003536	0.000820	0.026614	0.000584
8	MAXCO	0.723484	0.036610	0.619355	0.800000	0.722581	0.053775	0.003782	0.036501	0.000535

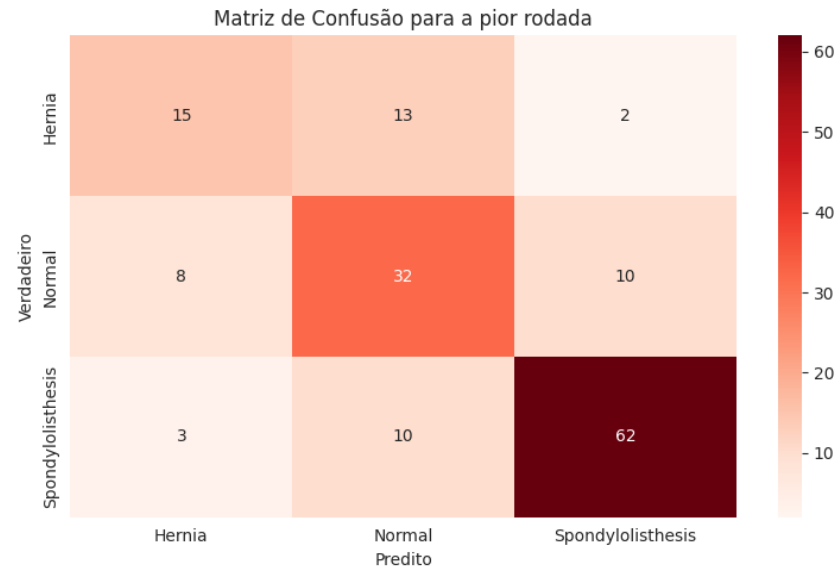
```
--- KNN (m=0.50) ---
Melhor rodada (#47) Acurácia: 0.8387
Pior rodada (#59) Acurácia: 0.6968
```



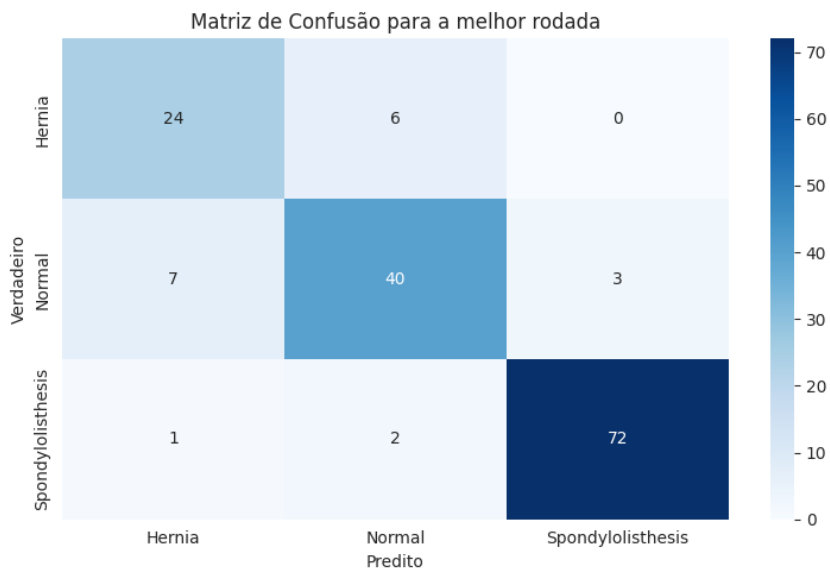


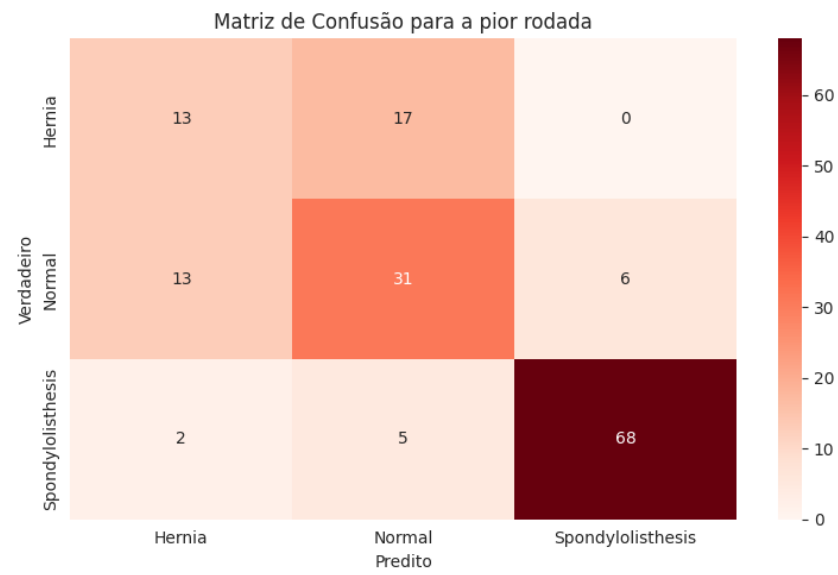

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.81      0.70      0.75        30
           Normal       0.73      0.82      0.77        50
Spondylolisthesis       0.93      0.91      0.92        75

         accuracy                           0.84       155
        macro avg       0.82      0.81      0.81       155
     weighted avg       0.84      0.84      0.84       155

--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.8645
Pior rodada (#59) Acurácia: 0.7032
```



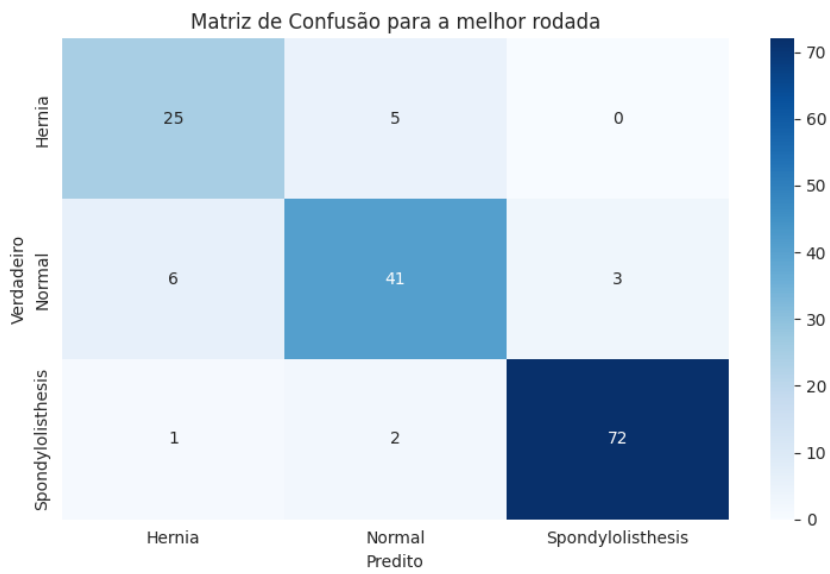


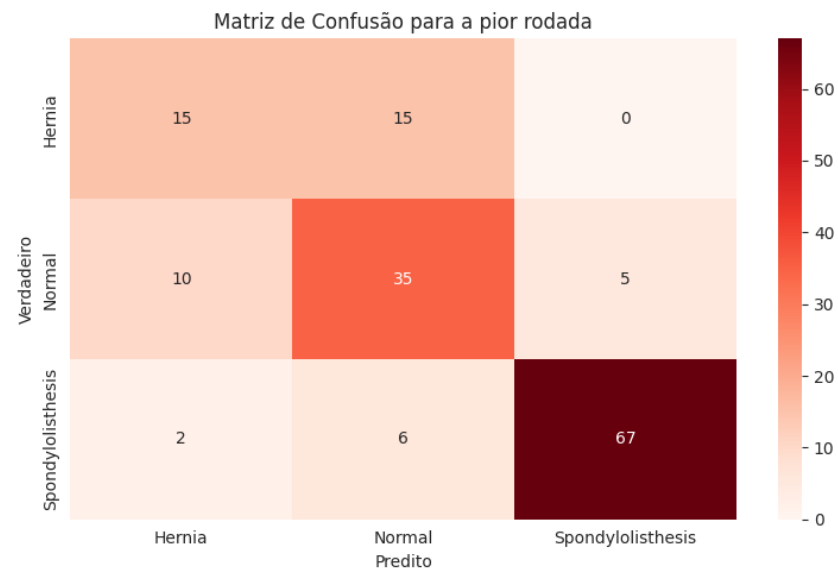

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.80      0.67      0.73        30
           Normal       0.78      0.84      0.81        50
Spondylolisthesis       0.95      0.96      0.95        75

         accuracy                           0.86       155
        macro avg       0.84      0.82      0.83       155
     weighted avg       0.86      0.86      0.86       155

--- KNN (m=1.00) ---
Melhor rodada (#56) Acurácia: 0.8774
Pior rodada (#43) Acurácia: 0.7226
```



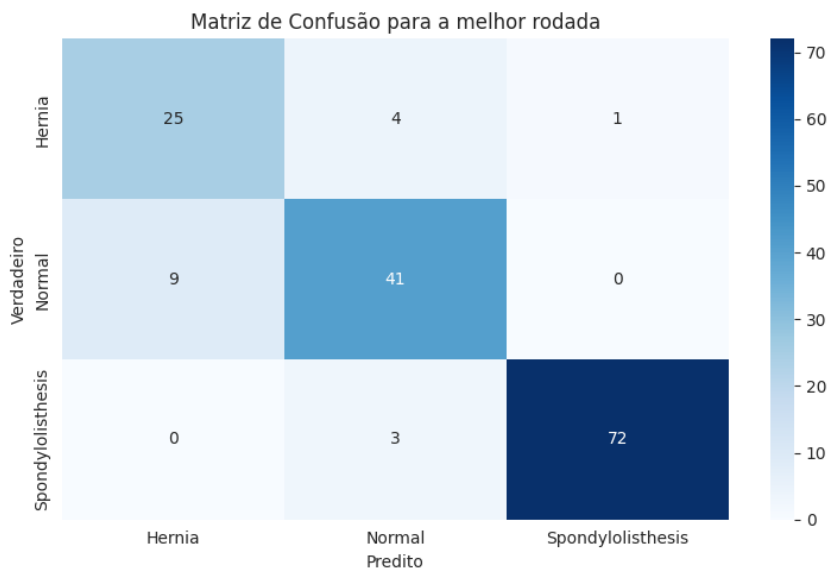


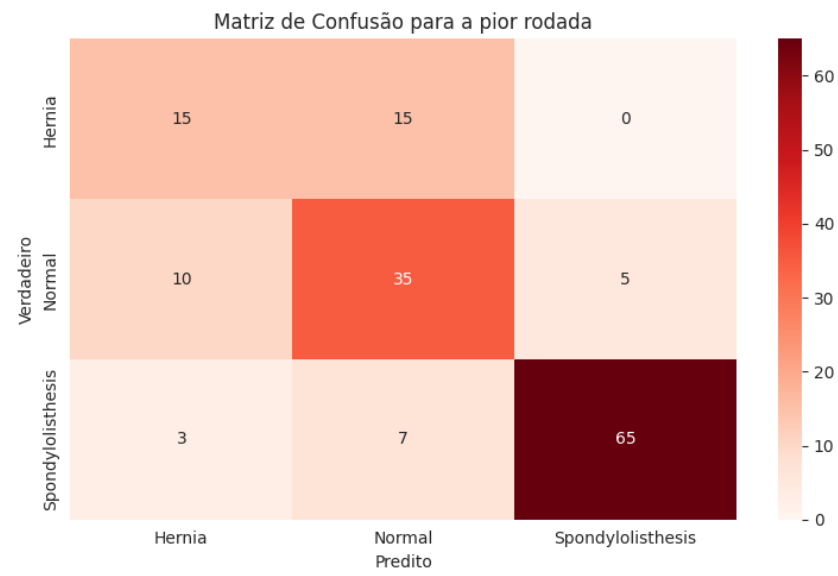

```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.75      0.80      0.77        30
           Normal       0.83      0.80      0.82        50
Spondylolisthesis       0.96      0.96      0.96        75

         accuracy                           0.88       155
        macro avg       0.85      0.85      0.85       155
     weighted avg       0.88      0.88      0.88       155

--- KNN (m=1.50) ---
Melhor rodada (#56) Acurácia: 0.8903
Pior rodada (#11) Acurácia: 0.7548
```



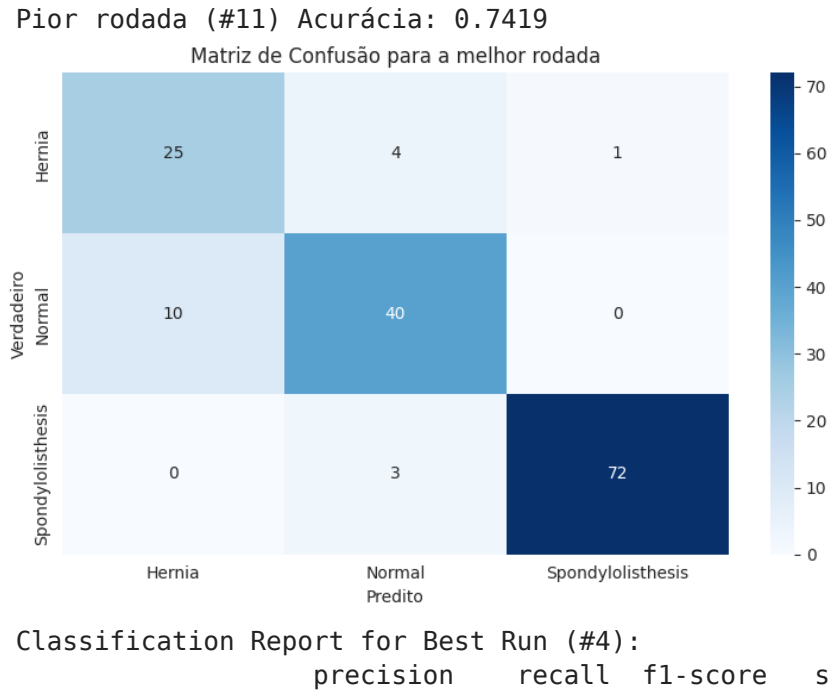


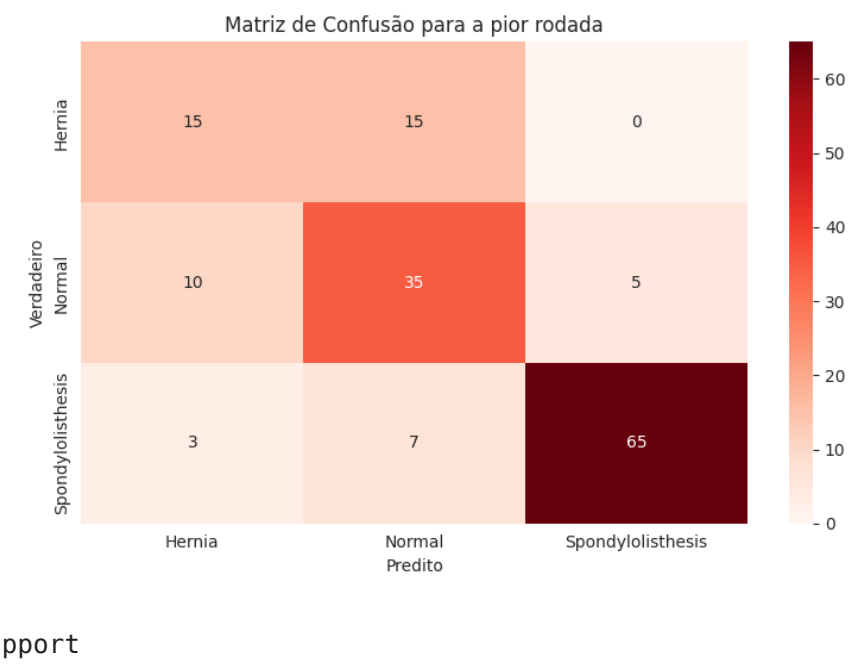
```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.78      0.83      0.81        30
           Normal       0.85      0.82      0.84        50
Spondylolisthesis       0.96      0.96      0.96        75

         accuracy                           0.89       155
        macro avg       0.87      0.87      0.87       155
     weighted avg       0.89      0.89      0.89       155

--- KNN (m=2.00) ---
Melhor rodada (#4) Acurácia: 0.8903
Pior rodada (#11) Acurácia: 0.7419
```



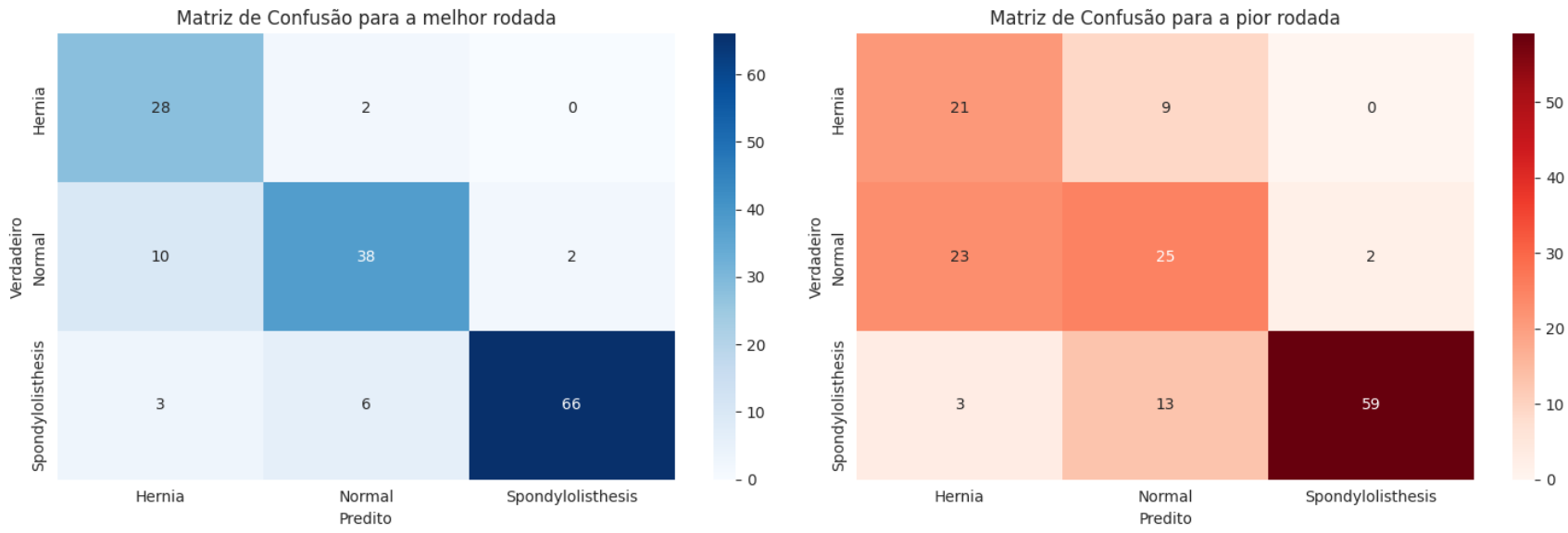



```
Classification Report for Best Run (#4):
                   precision    recall  f1-score   support

           Hernia       0.74      0.83      0.78        30
           Normal       0.85      0.82      0.84        50
Spondylolisthesis       0.99      0.96      0.97        75

         accuracy                           0.89       155
        macro avg       0.86      0.87      0.86       155
     weighted avg       0.90      0.89      0.89       155

--- KNN (m=2.50) ---
Melhor rodada (#4) Acurácia: 0.8839
Pior rodada (#11) Acurácia: 0.7419
```

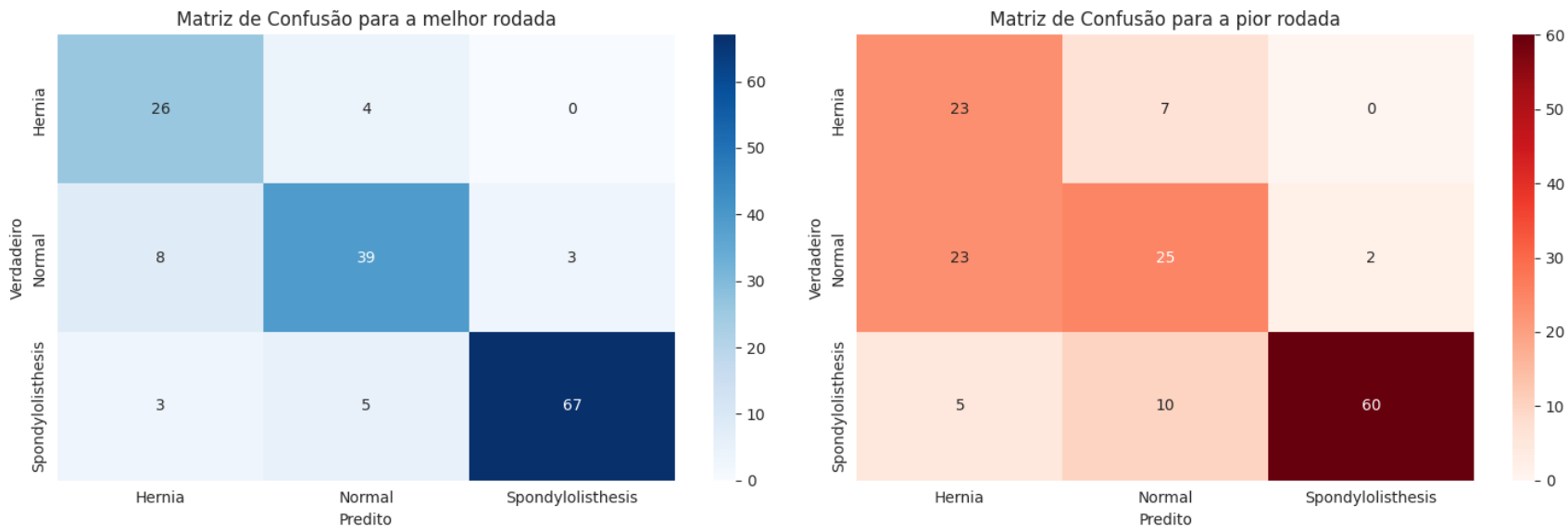




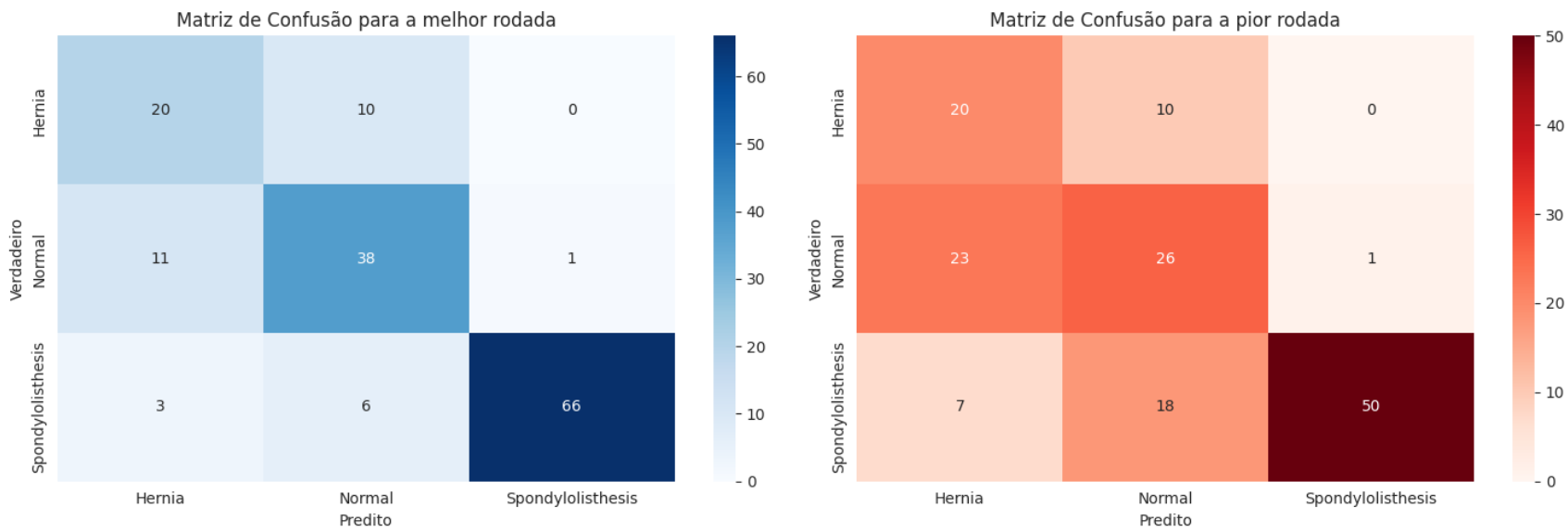

```
Classification Report for Best Run (#4):
                   precision    recall  f1-score   support

           Hernia       0.71      0.83      0.77        30
           Normal       0.85      0.80      0.82        50
Spondylolisthesis       0.99      0.96      0.97        75

         accuracy                           0.88       155
        macro avg       0.85      0.86      0.86       155
     weighted avg       0.89      0.88      0.89       155

--- MDC ---
Melhor rodada (#24) Acurácia: 0.8516
Pior rodada (#94) Acurácia: 0.6774
```




```
Classification Report for Best Run (#24):
                   precision    recall  f1-score   support

           Hernia       0.68      0.93      0.79        30
           Normal       0.83      0.76      0.79        50
Spondylolisthesis       0.97      0.88      0.92        75

         accuracy                           0.85       155
        macro avg       0.83      0.86      0.83       155
     weighted avg       0.87      0.85      0.85       155

--- MDC Robusto ---
Melhor rodada (#56) Acurácia: 0.8516
Pior rodada (#54) Acurácia: 0.6968
```



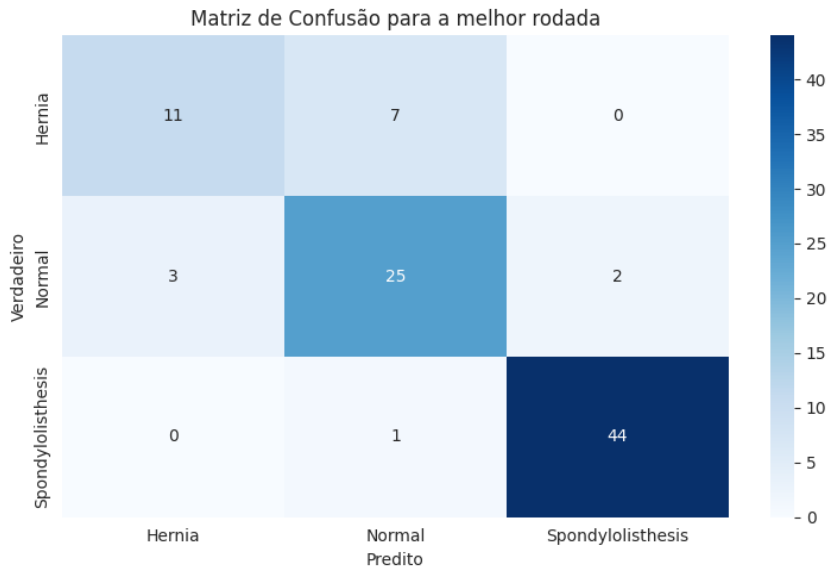

```
Classification Report for Best Run (#56):
                   precision    recall  f1-score   support

           Hernia       0.70      0.87      0.78        30
           Normal       0.81      0.78      0.80        50
Spondylolisthesis       0.96      0.89      0.92        75

         accuracy                           0.85       155
        macro avg       0.82      0.85      0.83       155
     weighted avg       0.86      0.85      0.85       155

--- MAXCO ---
Melhor rodada (#83) Acurácia: 0.8000
Pior rodada (#54) Acurácia: 0.6194
```



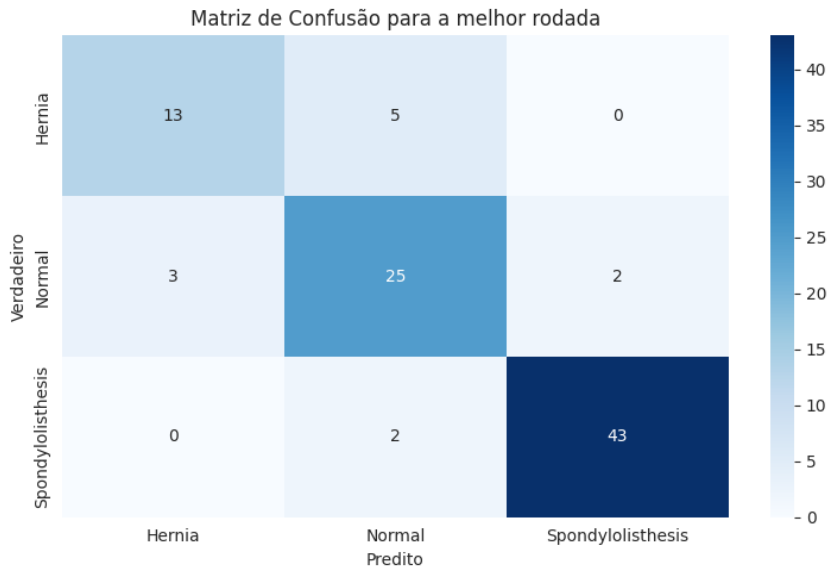
```
Classification Report for Best Run (#83):
                   precision    recall  f1-score   support

           Hernia       0.59      0.67      0.62        30
           Normal       0.70      0.76      0.73        50
Spondylolisthesis       0.99      0.88      0.93        75

         accuracy                           0.80       155
        macro avg       0.76      0.77      0.76       155
     weighted avg       0.82      0.80      0.81       155

====================================================================
   Rodando Experimento para Divisão: 30/70
====================================================================
```

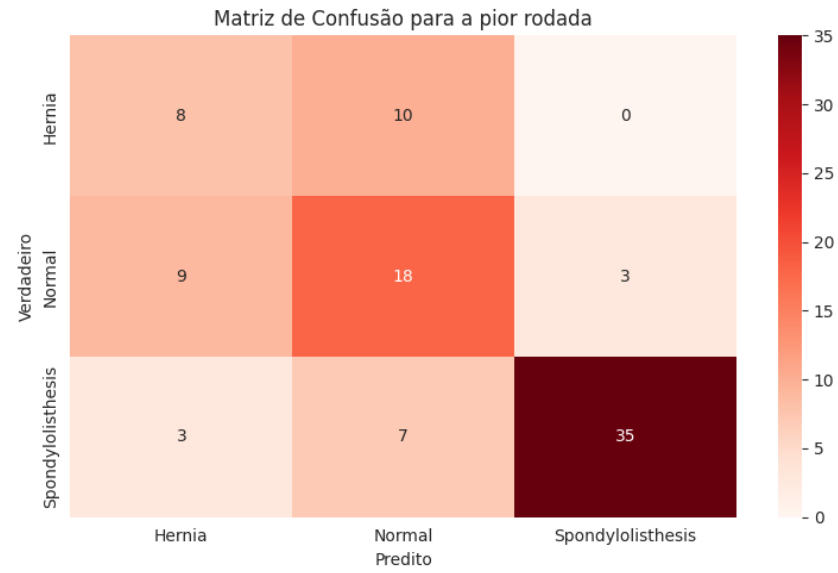
	Classifier	Acuracia Media	Std Acuracia	Min Acuracia	Max Acuracia	Median Acuracia	Tempo de Treino Medio (ms)	Tempo de Treino Std (ms)	Tempo de Teste Medio (ms)	Tempo de Teste Std (ms)
5	KNN (m=2.50)	0.814409	0.034463	0.752688	0.903226	0.806452	0.002444	0.000271	0.045416	0.001256
4	KNN (m=2.00)	0.812151	0.035352	0.741935	0.892473	0.806452	0.002487	0.000260	0.027873	0.000374
3	KNN (m=1.50)	0.807849	0.033923	0.731183	0.903226	0.806452	0.002484	0.000244	0.045337	0.000568
2	KNN (m=1.00)	0.805699	0.037473	0.709677	0.913978	0.806452	0.002611	0.000922	0.027982	0.000496
7	MDC Robusto	0.786667	0.037364	0.698925	0.870968	0.795699	0.135489	0.006825	0.005606	0.000078
1	KNN (m=0.67)	0.780860	0.037918	0.655914	0.870968	0.784946	0.002606	0.000289	0.045415	0.000539
6	MDC	0.765914	0.039679	0.655914	0.870968	0.763441	0.067055	0.005847	0.005547	0.000085
0	KNN (m=0.50)	0.765699	0.038927	0.655914	0.860215	0.774194	0.003583	0.000848	0.029077	0.000394
8	MAXCO	0.722043	0.046158	0.580645	0.849462	0.720430	0.055449	0.001666	0.036626	0.000462

```
--- KNN (m=0.50) ---
Melhor rodada (#98) Acurácia: 0.8602
Pior rodada (#57) Acurácia: 0.6559
```

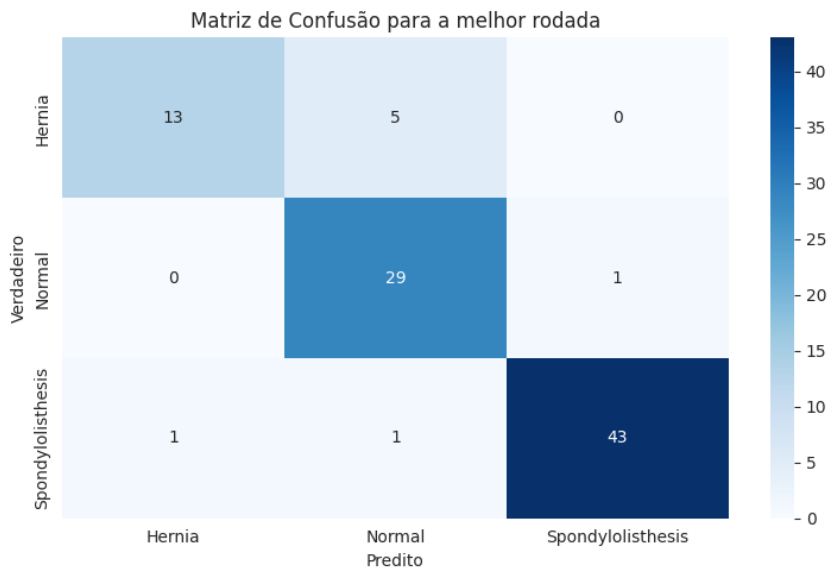


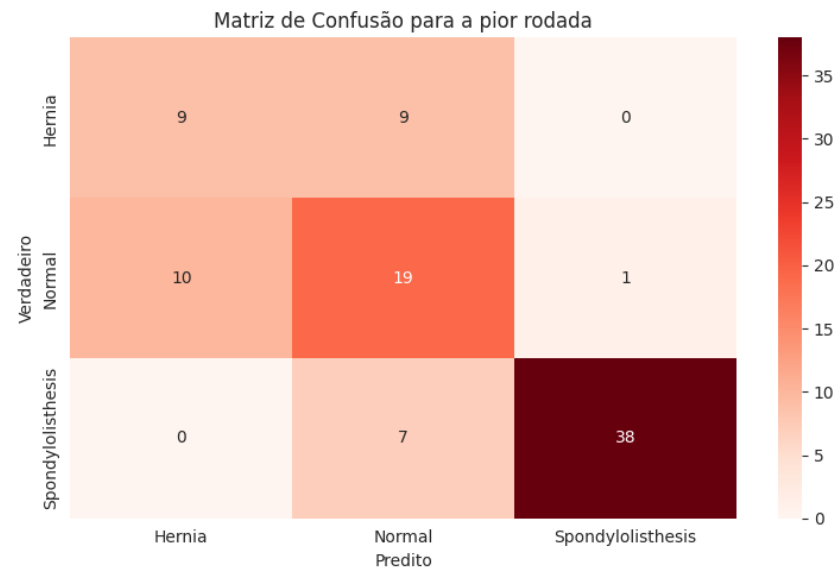
```
Classification Report for Best Run (#98):
                   precision    recall  f1-score   support

           Hernia       0.79      0.61      0.69        18
           Normal       0.76      0.83      0.79        30
Spondylolisthesis       0.96      0.98      0.97        45

         accuracy                           0.86        93
        macro avg       0.83      0.81      0.82        93
     weighted avg       0.86      0.86      0.86        93

--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.8710
Pior rodada (#57) Acurácia: 0.6559
```



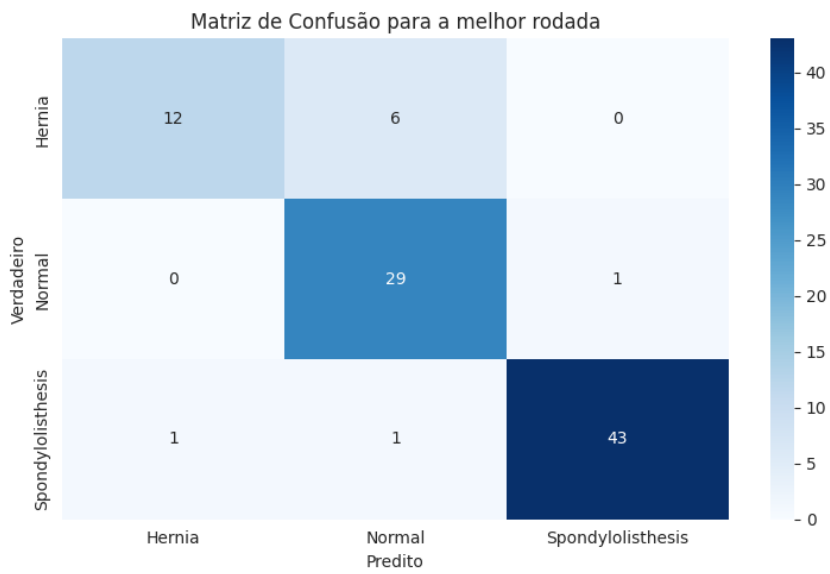


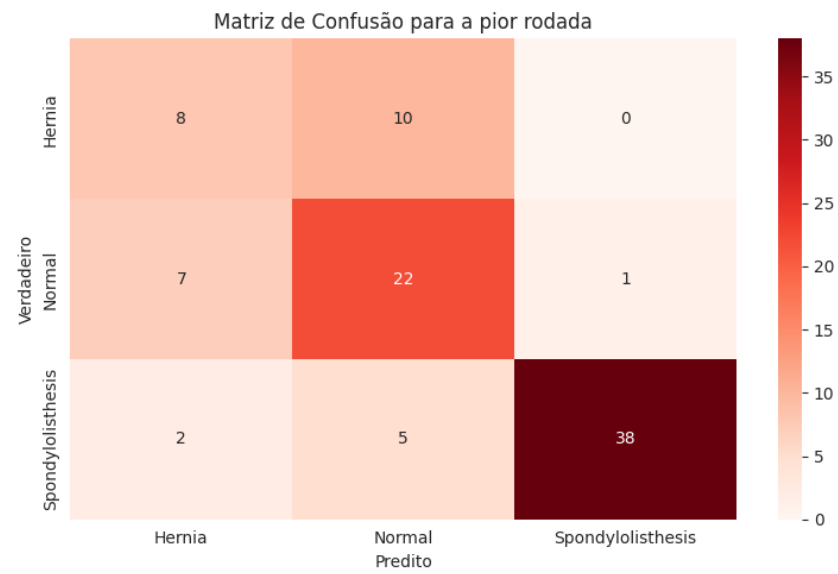

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.81      0.72      0.76        18
           Normal       0.78      0.83      0.81        30
Spondylolisthesis       0.96      0.96      0.96        45

         accuracy                           0.87        93
        macro avg       0.85      0.84      0.84        93
     weighted avg       0.87      0.87      0.87        93

--- KNN (m=1.00) ---
Melhor rodada (#65) Acurácia: 0.9140
Pior rodada (#12) Acurácia: 0.7097
```



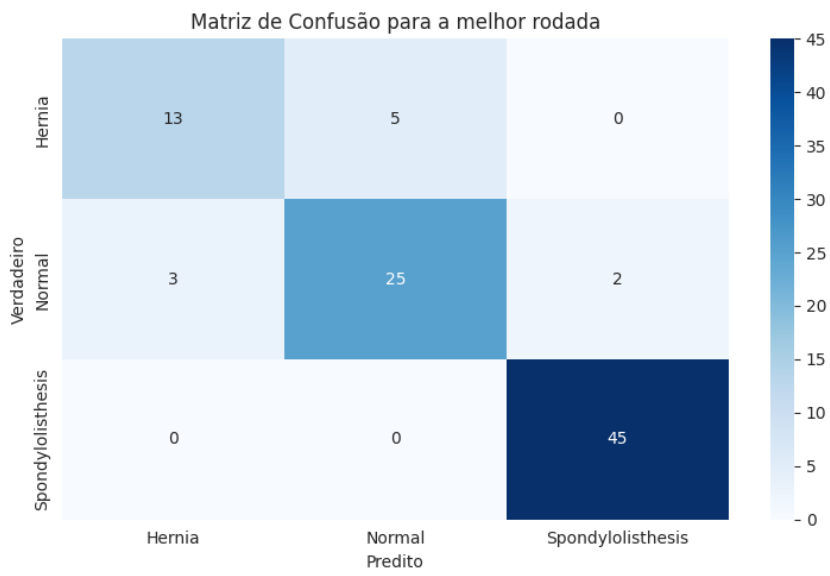


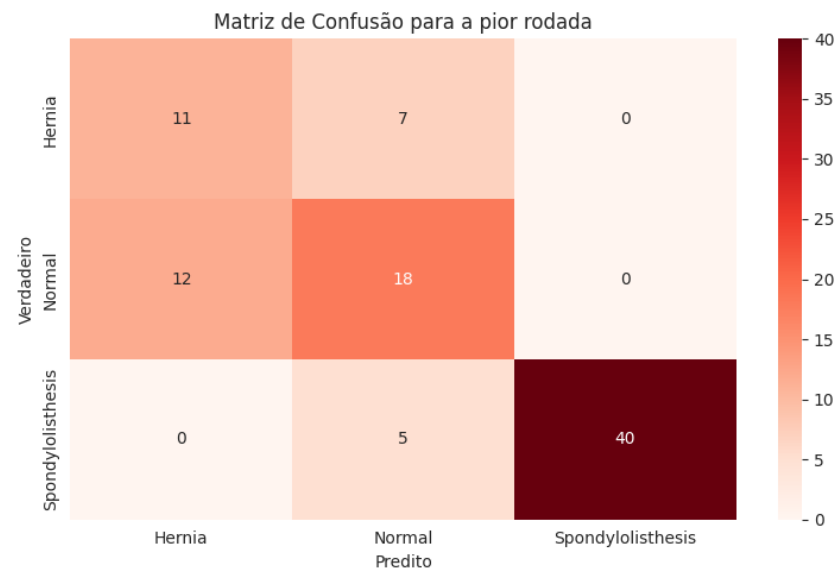

```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.93      0.72      0.81        18
           Normal       0.83      0.97      0.89        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.91        93
        macro avg       0.91      0.88      0.89        93
     weighted avg       0.92      0.91      0.91        93

--- KNN (m=1.50) ---
Melhor rodada (#65) Acurácia: 0.9032
Pior rodada (#29) Acurácia: 0.7312
```



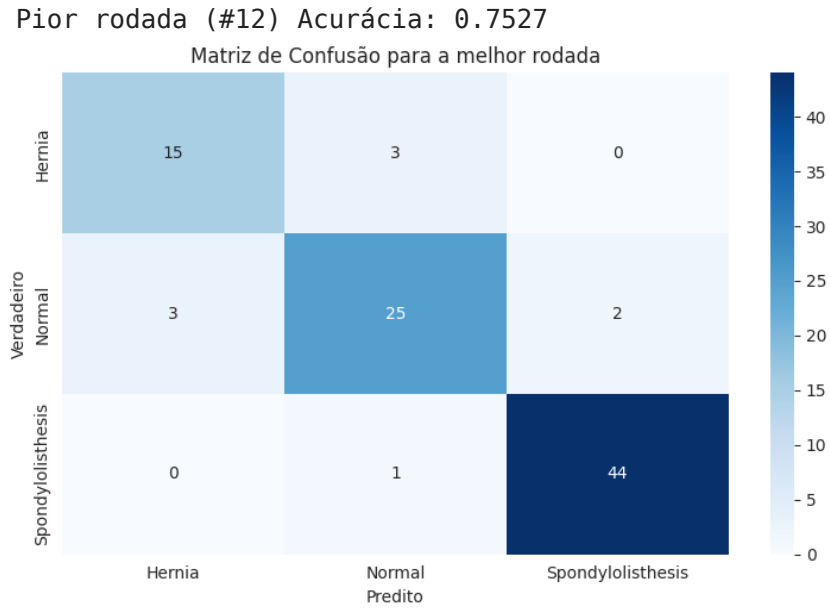


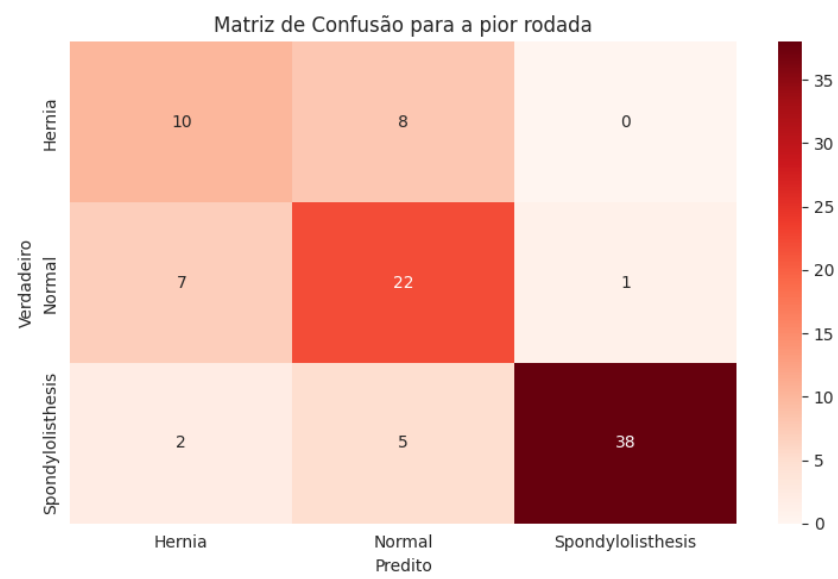
```
Classification Report for Best Run (#65):
                   precision    recall  f1-score   support

           Hernia       0.92      0.67      0.77        18
           Normal       0.81      0.97      0.88        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.90        93
        macro avg       0.90      0.86      0.87        93
     weighted avg       0.91      0.90      0.90        93

--- KNN (m=2.00) ---
Melhor rodada (#23) Acurácia: 0.8925
Pior rodada (#3) Acurácia: 0.7419
```



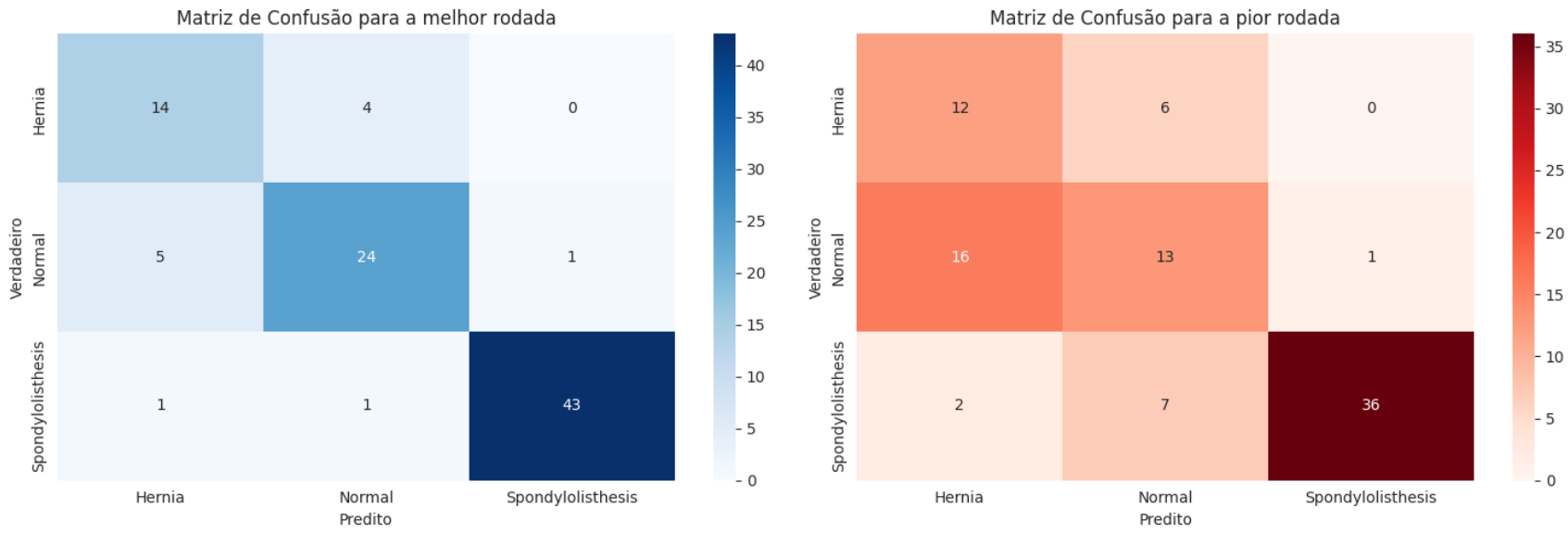



```
Classification Report for Best Run (#23):
                   precision    recall  f1-score   support

           Hernia       0.81      0.72      0.76        18
           Normal       0.83      0.83      0.83        30
Spondylolisthesis       0.96      1.00      0.98        45

         accuracy                           0.89        93
        macro avg       0.87      0.85      0.86        93
     weighted avg       0.89      0.89      0.89        93

--- KNN (m=2.50) ---
Melhor rodada (#99) Acurácia: 0.9032
Pior rodada (#12) Acurácia: 0.7527
```

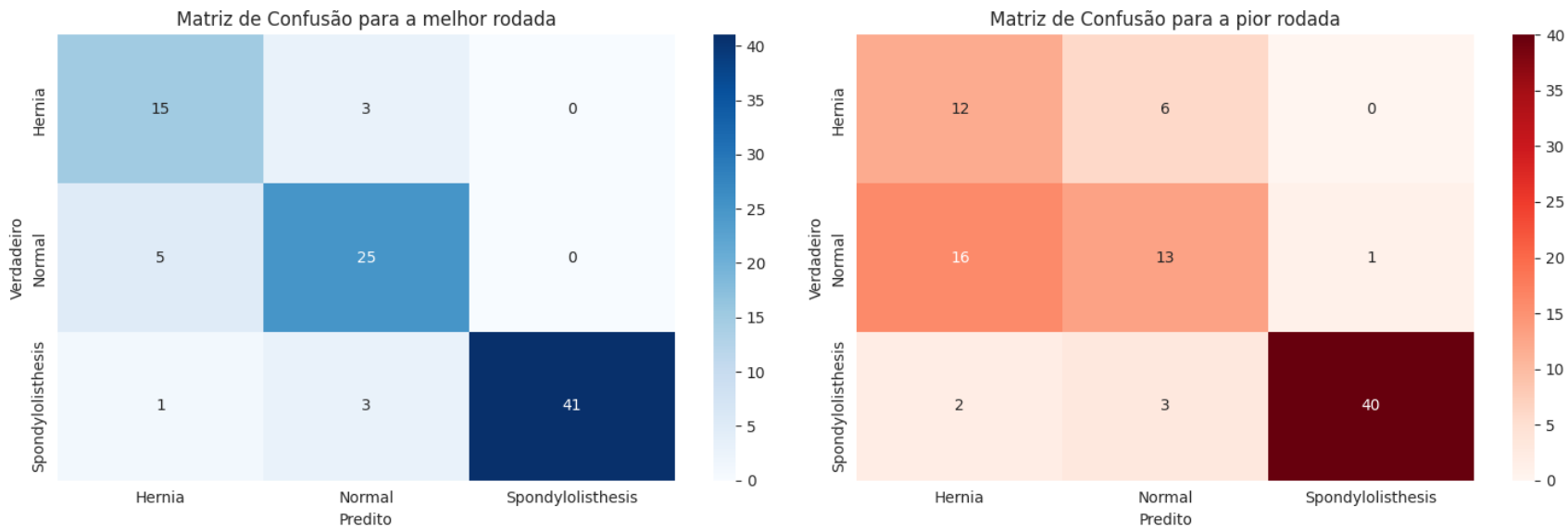




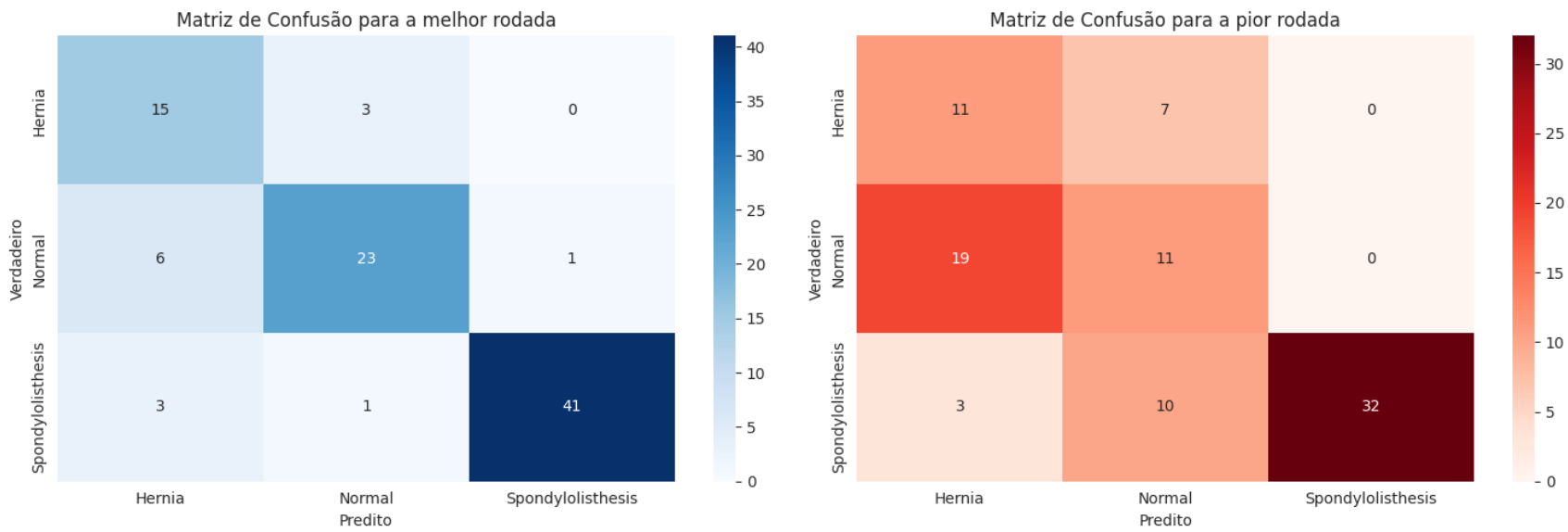

```
Classification Report for Best Run (#99):
                   precision    recall  f1-score   support

           Hernia       0.83      0.83      0.83        18
           Normal       0.86      0.83      0.85        30
Spondylolisthesis       0.96      0.98      0.97        45

         accuracy                           0.90        93
        macro avg       0.88      0.88      0.88        93
     weighted avg       0.90      0.90      0.90        93

--- MDC ---
Melhor rodada (#28) Acurácia: 0.8710
Pior rodada (#3) Acurácia: 0.6559
```




```
Classification Report for Best Run (#28):
                   precision    recall  f1-score   support

           Hernia       0.70      0.78      0.74        18
           Normal       0.83      0.80      0.81        30
Spondylolisthesis       0.98      0.96      0.97        45

         accuracy                           0.87        93
        macro avg       0.83      0.84      0.84        93
     weighted avg       0.88      0.87      0.87        93

--- MDC Robusto ---
Melhor rodada (#80) Acurácia: 0.8710
Pior rodada (#3) Acurácia: 0.6989
```



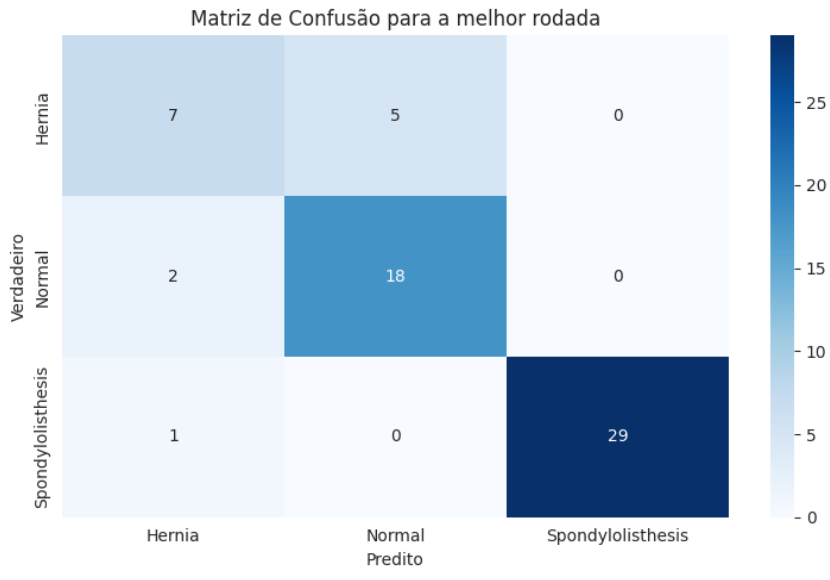

```
Classification Report for Best Run (#80):
                   precision    recall  f1-score   support

           Hernia       0.71      0.83      0.77        18
           Normal       0.81      0.83      0.82        30
Spondylolisthesis       1.00      0.91      0.95        45

         accuracy                           0.87        93
        macro avg       0.84      0.86      0.85        93
     weighted avg       0.88      0.87      0.87        93

--- MAXCO ---
Melhor rodada (#79) Acurácia: 0.8495
Pior rodada (#22) Acurácia: 0.5806
```

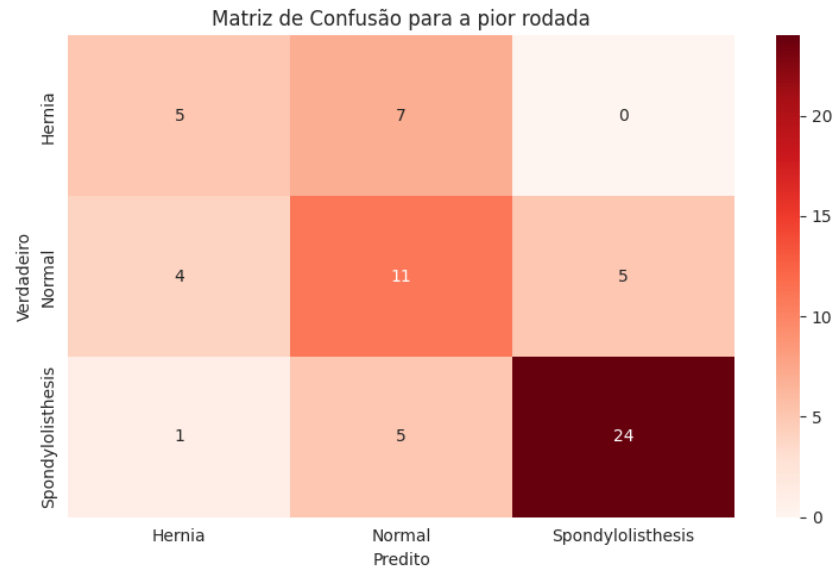


```
Classification Report for Best Run (#79):
                   precision    recall  f1-score   support

           Hernia       0.62      0.83      0.71        18
           Normal       0.85      0.77      0.81        30
Spondylolisthesis       0.98      0.91      0.94        45

         accuracy                           0.85        93
        macro avg       0.82      0.84      0.82        93
     weighted avg       0.87      0.85      0.85        93

==================================================================
   Rodando Experimento para Divisão: 20/80
==================================================================
```

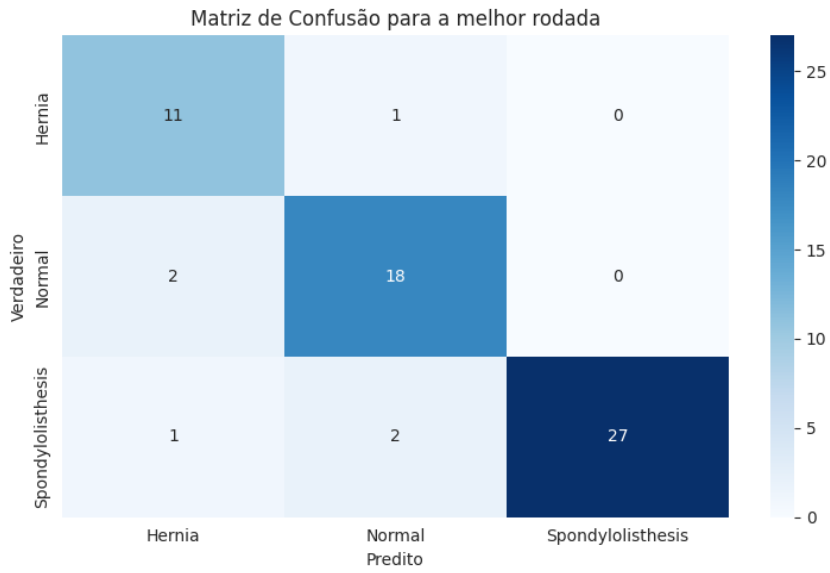
	Classifier	Acuracia Media	Std Acuracia	Min Acuracia	Max Acuracia	Median Acuracia	Tempo de Treino Medio (ms)	Tempo de Treino Std (ms)	Tempo de Teste Medio (ms)	Tempo de Teste Std (ms)
5	KNN (m=2.50)	0.823871	0.042040	0.725806	0.919355	0.822581	0.002418	0.000290	0.048983	0.001958
4	KNN (m=2.00)	0.820968	0.042275	0.693548	0.903226	0.822581	0.002484	0.000425	0.029160	0.001722
3	KNN (m=1.50)	0.816129	0.041624	0.725806	0.903226	0.806452	0.002520	0.000662	0.049229	0.003002
2	KNN (m=1.00)	0.815806	0.041196	0.709677	0.935484	0.814516	0.002644	0.000795	0.029462	0.002023
7	MDC Robusto	0.786129	0.045024	0.677419	0.870968	0.782258	0.137849	0.006308	0.005544	0.000072
1	KNN (m=0.67)	0.784516	0.048949	0.629032	0.903226	0.790323	0.002606	0.000291	0.049453	0.002997
0	KNN (m=0.50)	0.765484	0.045243	0.645161	0.870968	0.758065	0.003543	0.000603	0.030511	0.001845
6	MDC	0.762903	0.046495	0.661290	0.887097	0.774194	0.067484	0.004221	0.005564	0.000085
8	MAXCO	0.722097	0.060084	0.564516	0.887097	0.709677	0.055976	0.002456	0.036869	0.000973

```
--- KNN (m=0.50) ---
Melhor rodada (#92) Acurácia: 0.8710
Pior rodada (#30) Acurácia: 0.6452
```

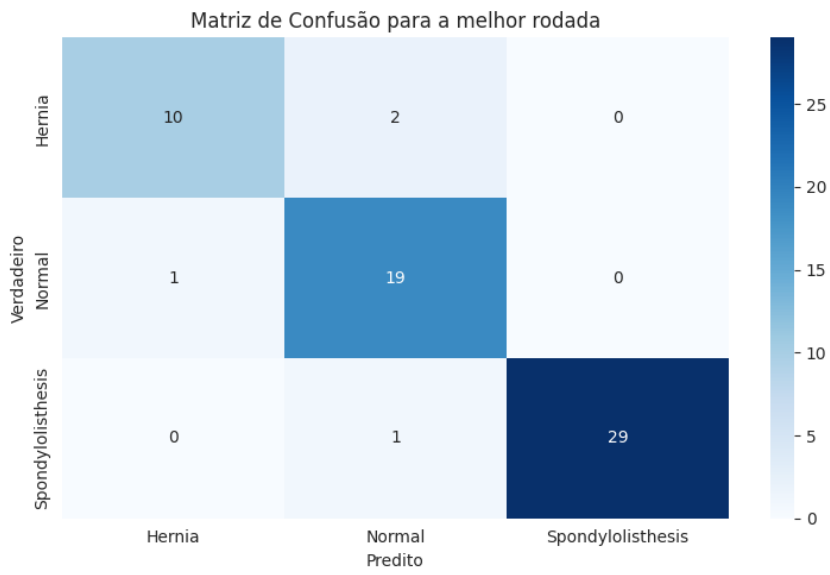


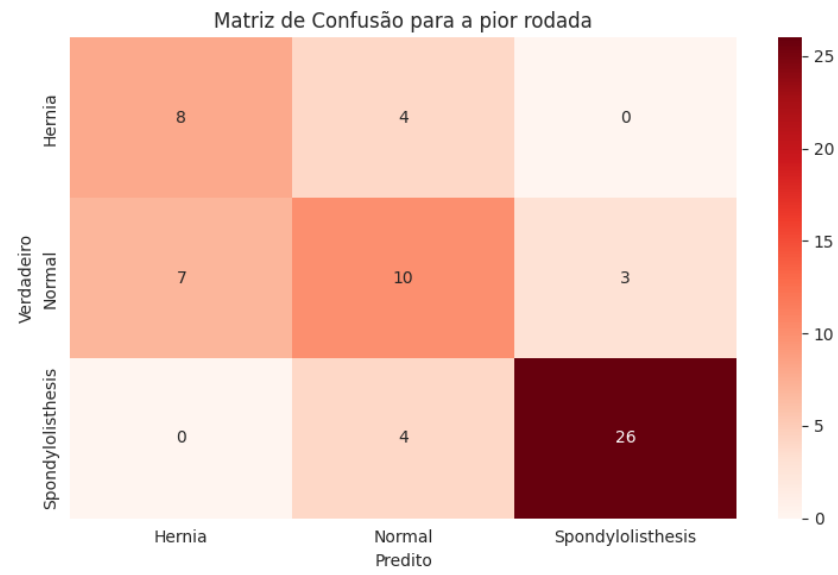



```
Classification Report for Best Run (#92):
                   precision    recall  f1-score   support

           Hernia       0.70      0.58      0.64        12
           Normal       0.78      0.90      0.84        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.87        62
        macro avg       0.83      0.82      0.82        62
     weighted avg       0.87      0.87      0.87        62

--- KNN (m=0.67) ---
Melhor rodada (#47) Acurácia: 0.9032
Pior rodada (#30) Acurácia: 0.6290
```

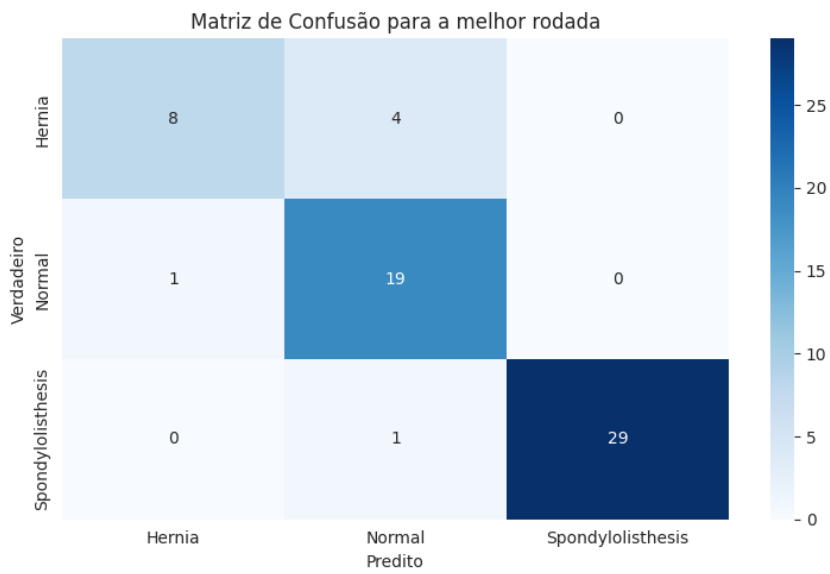


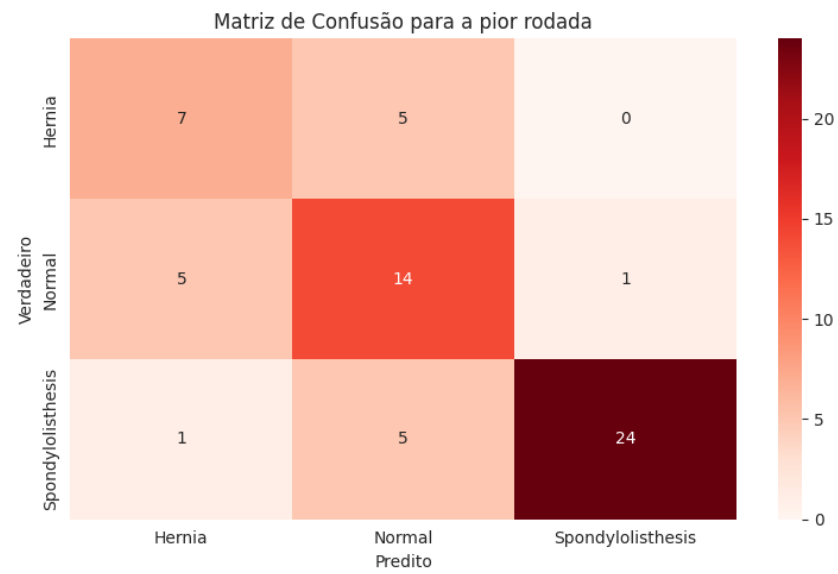
```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.79      0.92      0.85        12
           Normal       0.86      0.90      0.88        20
Spondylolisthesis       1.00      0.90      0.95        30

         accuracy                           0.90        62
        macro avg       0.88      0.91      0.89        62
     weighted avg       0.91      0.90      0.91        62

--- KNN (m=1.00) ---
Melhor rodada (#47) Acurácia: 0.9355
Pior rodada (#96) Acurácia: 0.7097
```



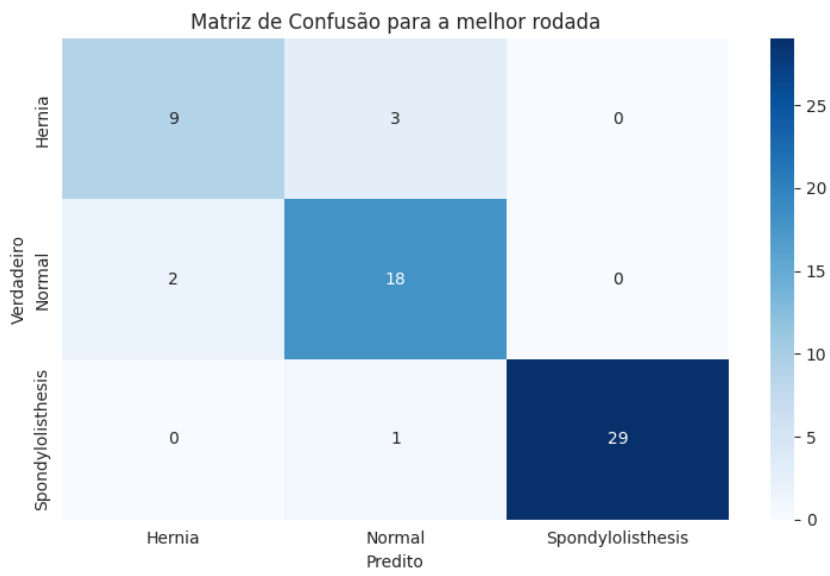


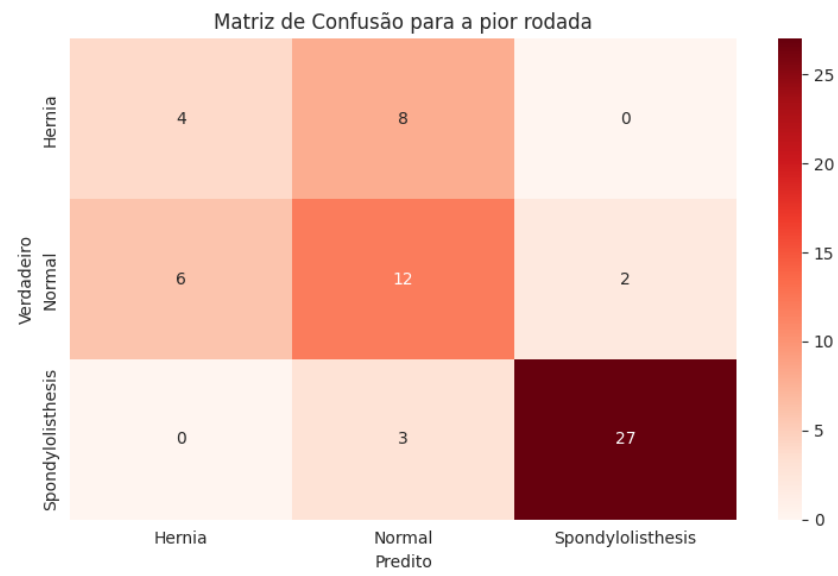

```
Classification Report for Best Run (#47):
                   precision    recall  f1-score   support

           Hernia       0.91      0.83      0.87        12
           Normal       0.86      0.95      0.90        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.94        62
        macro avg       0.92      0.92      0.92        62
     weighted avg       0.94      0.94      0.94        62

--- KNN (m=1.50) ---
Melhor rodada (#55) Acurácia: 0.9032
Pior rodada (#29) Acurácia: 0.7258
```



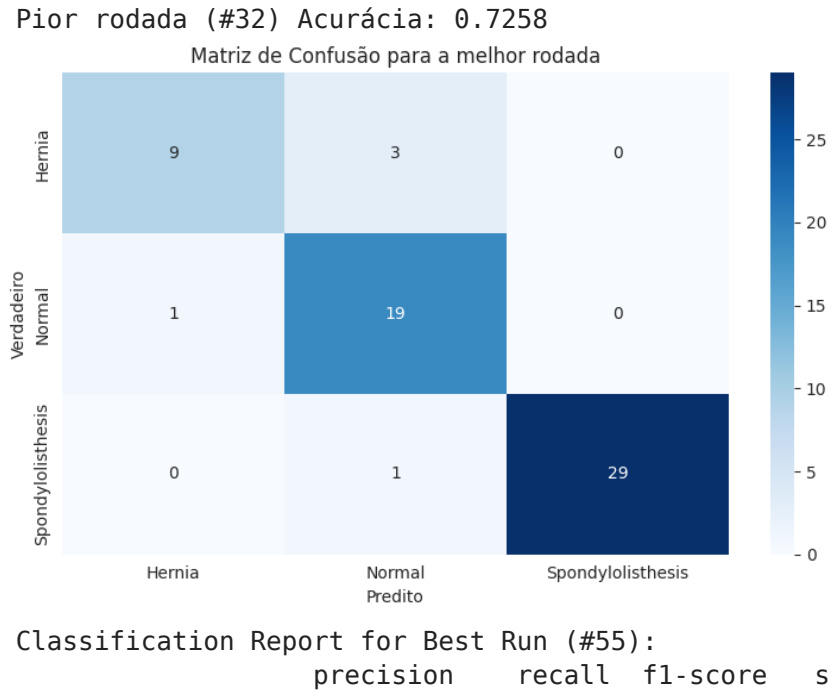


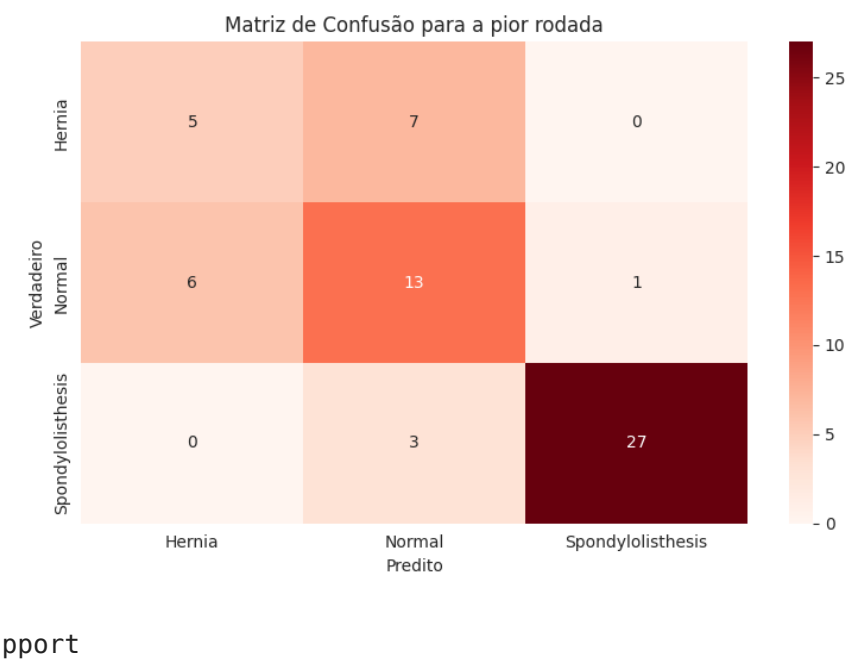
```
Classification Report for Best Run (#55):
                   precision    recall  f1-score   support

           Hernia       0.89      0.67      0.76        12
           Normal       0.79      0.95      0.86        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.90        62
        macro avg       0.89      0.86      0.87        62
     weighted avg       0.91      0.90      0.90        62

--- KNN (m=2.00) ---
Melhor rodada (#4) Acurácia: 0.9032
Pior rodada (#32) Acurácia: 0.6935
```



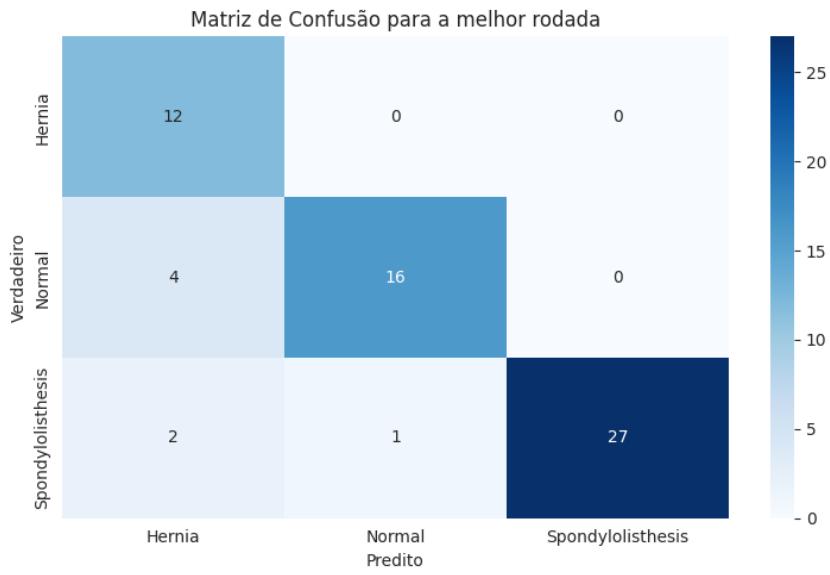


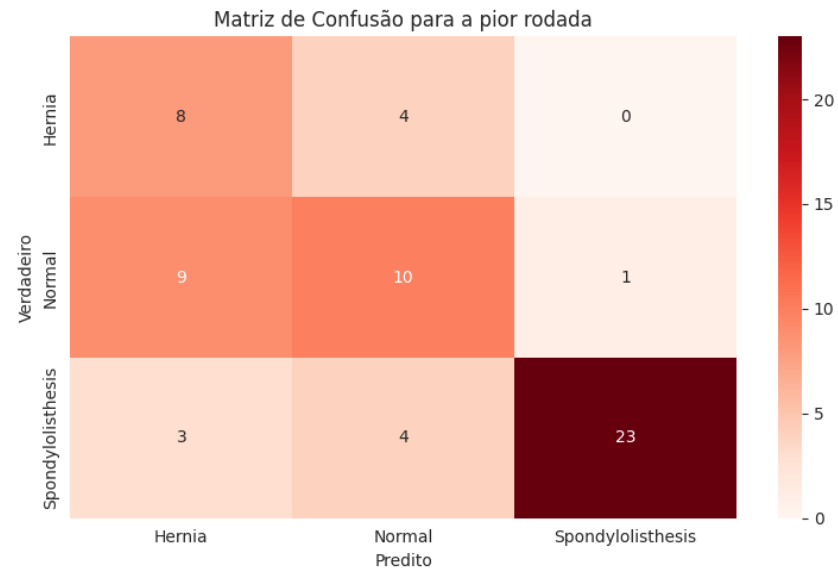

```
Classification Report for Best Run (#4):
                   precision    recall  f1-score   support

           Hernia       0.82      0.75      0.78        12
           Normal       0.82      0.90      0.86        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.90        62
        macro avg       0.88      0.87      0.87        62
     weighted avg       0.91      0.90      0.90        62

--- KNN (m=2.50) ---
Melhor rodada (#55) Acurácia: 0.9194
Pior rodada (#32) Acurácia: 0.7258
```



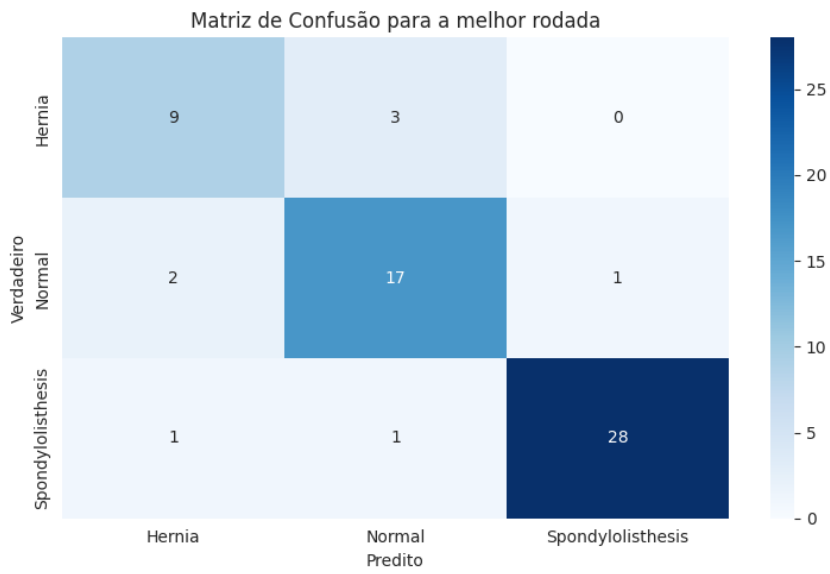


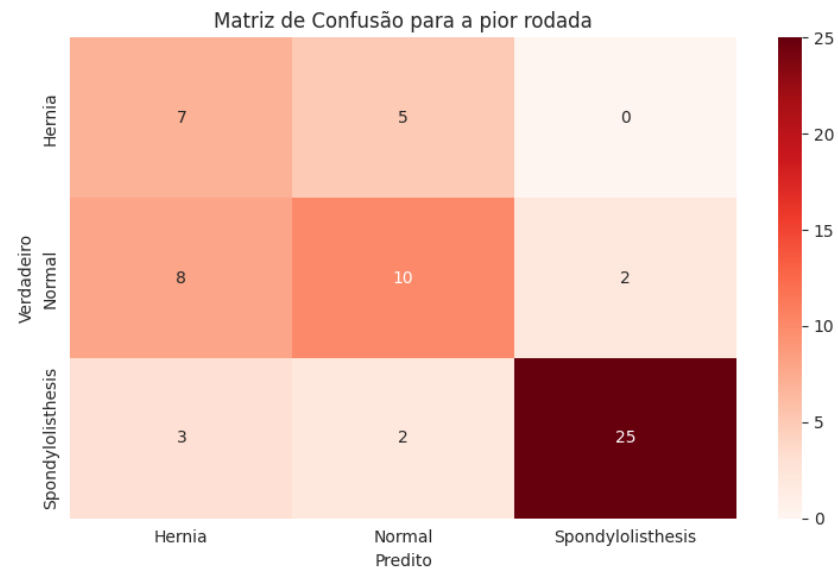

```
Classification Report for Best Run (#55):
                   precision    recall  f1-score   support

           Hernia       0.90      0.75      0.82        12
           Normal       0.83      0.95      0.88        20
Spondylolisthesis       1.00      0.97      0.98        30

         accuracy                           0.92        62
        macro avg       0.91      0.89      0.89        62
     weighted avg       0.92      0.92      0.92        62

--- MDC ---
Melhor rodada (#23) Acurácia: 0.8871
Pior rodada (#13) Acurácia: 0.6613
```



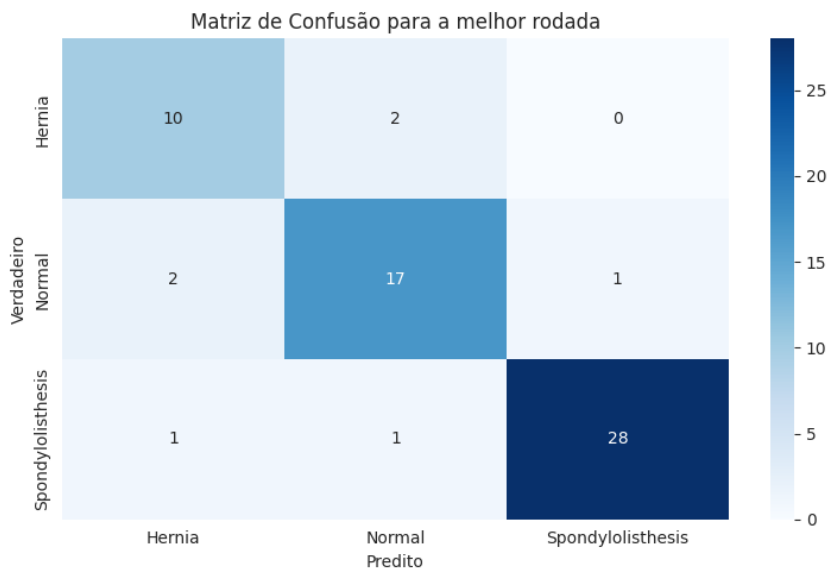


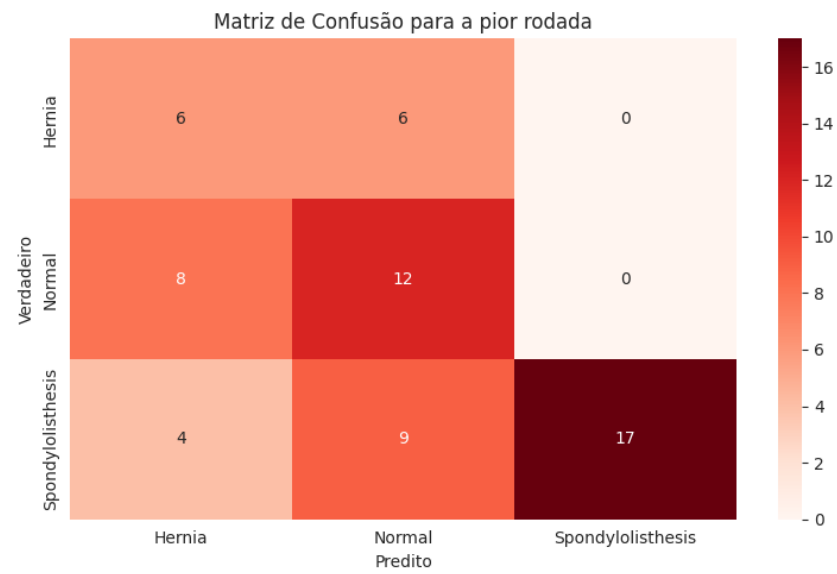

```
Classification Report for Best Run (#23):
                   precision    recall  f1-score   support

           Hernia       0.67      1.00      0.80        12
           Normal       0.94      0.80      0.86        20
Spondylolisthesis       1.00      0.90      0.95        30

         accuracy                           0.89        62
        macro avg       0.87      0.90      0.87        62
     weighted avg       0.92      0.89      0.89        62

--- MDC Robusto ---
Melhor rodada (#28) Acurácia: 0.8710
Pior rodada (#13) Acurácia: 0.6774
```






```
Classification Report for Best Run (#28):
                   precision    recall  f1-score   support

           Hernia       0.75      0.75      0.75        12
           Normal       0.81      0.85      0.83        20
Spondylolisthesis       0.97      0.93      0.95        30

         accuracy                           0.87        62
        macro avg       0.84      0.84      0.84        62
     weighted avg       0.87      0.87      0.87        62

--- MAXCO ---
Melhor rodada (#79) Acurácia: 0.8871
Pior rodada (#67) Acurácia: 0.5645
```





```
Classification Report for Best Run (#79):
                   precision    recall  f1-score   support

           Hernia       0.77      0.83      0.80        12
           Normal       0.85      0.85      0.85        20
Spondylolisthesis       0.97      0.93      0.95        30

         accuracy                           0.89        62
        macro avg       0.86      0.87      0.87        62
     weighted avg       0.89      0.89      0.89        62
```

Comparativo do desempenho dos algoritmos por proporção de dados de treinamento/teste

In [22]:
```
resultados_df.groupby(['Classifier', 'Train Ratio']).agg({
    'Acuracia Media': 'first',
    'Tempo de Treino Medio (ms)': 'first',
    'Tempo de Treino Std (ms)': 'first',
    'Tempo de Teste Medio (ms)': 'first',
    'Tempo de Teste Std (ms)': 'first'
}).rename(columns={
    'Acuracia Media': 'Acurácia',
    'Tempo de Treino Medio (ms)': 'Treino (ms)',
    'Tempo de Treino Std (ms)': 'Treino Std (ms)',
    'Tempo de Teste Medio (ms)': 'Teste (ms)',
    'Tempo de Teste Std (ms)': 'Teste Std (ms)'
})
```

Out[22]:

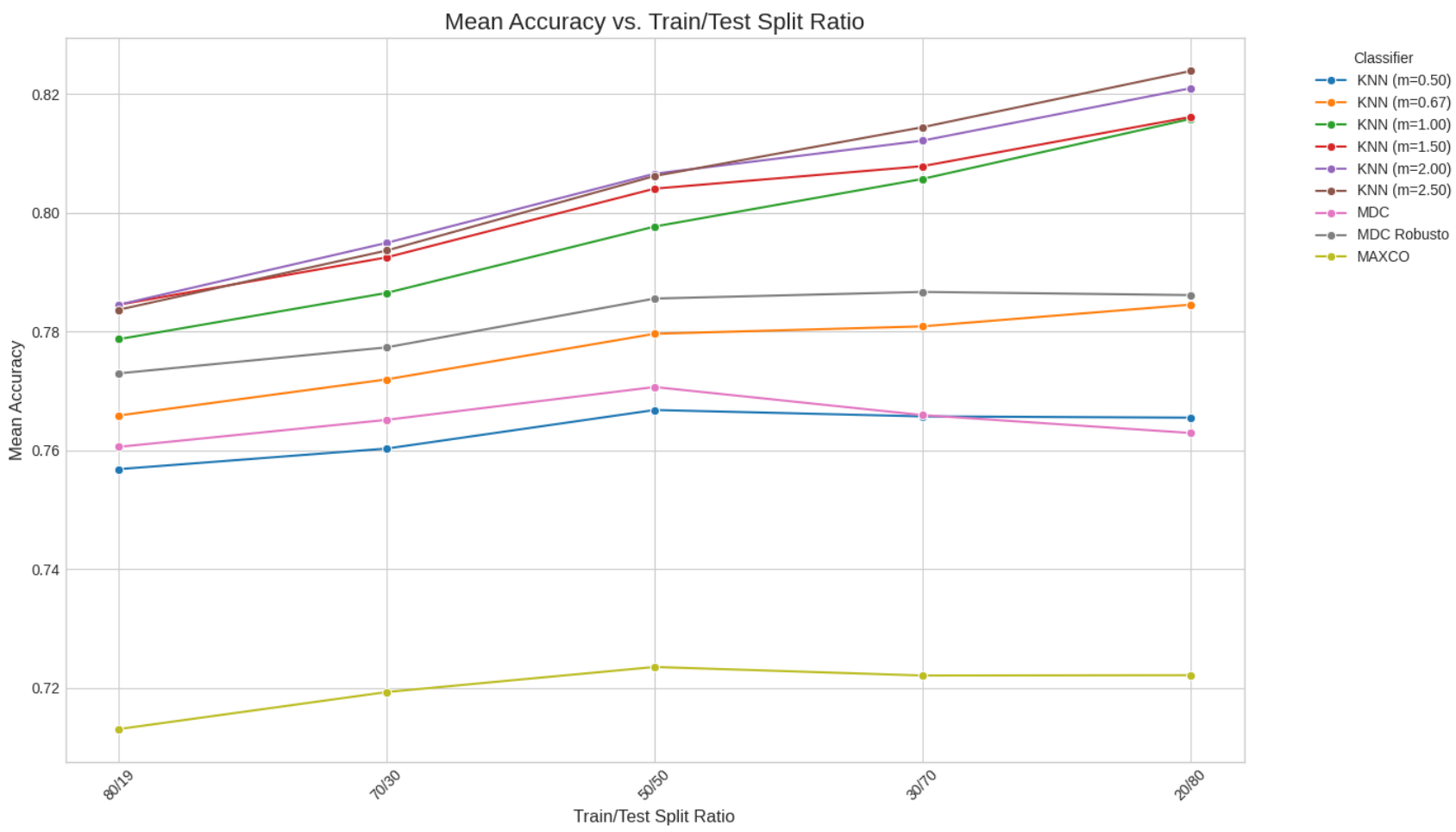
Classifier	Train Ratio	Acurácia	Treino (ms)	Treino Std (ms)	Teste (ms)	Teste Std (ms)
KNN (m=0.50)	**20**	0.765484	0.003543	0.000603	0.030511	0.001845
	30	0.765699	0.003583	0.000848	0.029077	0.000394
	50	0.766774	0.003536	0.000820	0.026614	0.000584
	70	0.760276	0.003555	0.000370	0.021965	0.000167
	80	0.756815	0.003786	0.000894	0.019871	0.000788
KNN (m=0.67)	**20**	0.784516	0.002606	0.000291	0.049453	0.002997
	30	0.780860	0.002606	0.000289	0.045415	0.000539
	50	0.779613	0.002668	0.000263	0.038400	0.000490
	70	0.771935	0.002639	0.000248	0.029436	0.000172
	80	0.765847	0.002780	0.000535	0.025164	0.001632
KNN (m=1.00)	**20**	0.815806	0.002644	0.000795	0.029462	0.002023
	30	0.805699	0.002611	0.000922	0.027982	0.000496
	50	0.797677	0.002632	0.000363	0.025804	0.000496
	70	0.786498	0.002542	0.000222	0.021773	0.000742
	80	0.778710	0.002716	0.000545	0.019745	0.001175
KNN (m=1.50)	**20**	0.816129	0.002520	0.000662	0.049229	0.003002
	30	0.807849	0.002484	0.000244	0.045337	0.000568
	50	0.804065	0.002539	0.000652	0.038489	0.001271
	70	0.792488	0.002525	0.000424	0.029365	0.000411
	80	0.784516	0.002599	0.000444	0.024999	0.001487
KNN (m=2.00)	**20**	0.820968	0.002484	0.000425	0.029160	0.001722
	30	0.812151	0.002487	0.000260	0.027873	0.000374
	50	0.806581	0.002549	0.000451	0.025654	0.000578
	70	0.794931	0.002558	0.000461	0.021382	0.000193
	80	0.784516	0.002663	0.000440	0.019282	0.000380
KNN (m=2.50)	**20**	0.823871	0.002418	0.000290	0.048983	0.001958
	30	0.814409	0.002444	0.000271	0.045416	0.001256
	50	0.806194	0.002532	0.000356	0.038477	0.001151
	70	0.793641	0.002451	0.000305	0.029387	0.000523
	80	0.783669	0.002627	0.000441	0.024817	0.000601
MAXCO	**20**	0.722097	0.055976	0.002456	0.036869	0.000973
	30	0.722043	0.055449	0.001666	0.036626	0.000462
	50	0.723484	0.053775	0.003782	0.036501	0.000535
	70	0.719263	0.051167	0.002975	0.036272	0.000226
	80	0.713024	0.049629	0.002506	0.036097	0.000444
MDC	**20**	0.762903	0.067484	0.004221	0.005564	0.000085
	30	0.765914	0.067055	0.005847	0.005547	0.000085
	50	0.770645	0.065167	0.005092	0.005471	0.000057
	70	0.765115	0.063791	0.006221	0.005452	0.000049
	80	0.760565	0.062957	0.004925	0.005377	0.000112
MDC Robusto	**20**	0.786129	0.137849	0.006308	0.005544	0.000072
	30	0.786667	0.135489	0.006825	0.005606	0.000078
	50	0.785548	0.129714	0.004919	0.005463	0.000057
	70	0.777327	0.124888	0.005391	0.005446	0.000060
	80	0.772944	0.122716	0.007636	0.005350	0.000045

5. OBS 3: Avaliação do Efeito da Separação Treino-Teste

In [17]:

```
split_analysis_data = []
for split_ratio, results in all_split_results.items():
    for name, runs in results.items():
        mean_accuracy = np.mean([r['accuracy'] for r in runs])
        split_analysis_data.append({'Split': split_ratio, 'Classifier': name, 'Mean Accuracy': mean_accurac

split_df = pd.DataFrame(split_analysis_data)
```

```
plt.style.use('seaborn-v0_8-whitegrid')
fig, ax = plt.subplots(figsize=(14, 8))
sns.lineplot(data=split_df, x='Split', y='Mean Accuracy', hue='Classifier', marker='o', ax=ax)
ax.set_title('Mean Accuracy vs. Train/Test Split Ratio', fontsize=16)
ax.set_xlabel('Train/Test Split Ratio', fontsize=12)
ax.set_ylabel('Mean Accuracy', fontsize=12)
ax.legend(title='Classifier', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [25]:

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd
from ucimlrepo import fetch_ucirepo

# Configurações
sns.set_style("whitegrid")
plt.rcParams.update({'font.size': 11})

# Carregamento (caso não esteja carregado)
vertebral_column = fetch_ucirepo(id=212)
df = vertebral_column.data.features.copy()
df['class'] = vertebral_column.data.targets.to_numpy().ravel()

# ---. Cálculo dos Centróides (Para argumentar sobre o MDC) ---
# Vamos calcular e exibir a distância entre os centróides para provar a separabilidade
print("\n--- Análise dos Centróides (Hipótese MDC) ---")
centroids = df.groupby('class').mean()
display(centroids)

# Dica de Mestre: Calcule a distância Euclidiana entre os centróides.
# Se a distância entre 'Hernia' e 'Normal' for pequena comparada à variância interna (dispersão),
# o MDC vai falhar miseravelmente nessas duas classes.
from scipy.spatial.distance import cdist
dists = pd.DataFrame(
    cdist(centroids, centroids, metric='euclidean'),
    index=centroids.index,
    columns=centroids.index
)
print("\nMatriz de Distâncias Euclidianas entre Centróides:")
display(dists.style.background_gradient(cmap='Blues'))
```

```
--- Análise dos Centróides (Hipótese MDC) ---
```

class	pelvic_incidence	pelvic_tilt	lumbar_lordosis_angle	sacral_slope	pelvic_radius	degree_spondylolisthesis
Hernia	47.638407	17.398795	35.463524	30.239612	116.474968	2.480251
Normal	51.685244	12.821414	43.542605	38.863830	123.890834	2.186572
Spondylolisthesis	71.514224	20.748038	64.110108	50.766186	114.518810	51.896687

```
Matriz de Distâncias Euclidianas entre Centróides:
```

class	Hernia	Normal	Spondylolisthesis
class			
Hernia	0.000000	15.233493	65.337966
Normal	15.233493	0.000000	59.830079
Spondylolisthesis	65.337966	59.830079	0.000000